Modeling Interest Rate Volatility In Ghana Using Markov-Switching (MS) Model

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ABSTRACT

This study applies Markov-Switching (MS) models to Ghana's key interest rate series covering January 2000 to August 2025. The goal is to identify latent volatility regimes, estimate regime-specific parameters, and interpret structural breaks within the context of macroeconomic and policy events. The analysis finds strong evidence of regime shifts, typically corresponding to major economic disruptions such as the global financial crisis, domestic energy and currency crises, COVID-19, and recent debt restructuring episodes. The MS framework provides superior interpretability compared to two-regime alternatives and offers useful insights for both policymakers and investors.

Key words: Markos- Switching, Interest rate, two-regime alternatives

1. INTRODUCTION

Interest rates such as the Monetary Policy Rate (MPR), lending rates, interbank rates, and average yields on 91- and 182-day Treasury bills are critical determinants of investment decisions. In Ghana's dynamic economic landscape, these rates often exhibit **regime-dependent behavior**, influenced by policy shifts, macroeconomic shocks, or global developments. Traditional single-regime models fail to capture sudden structural shifts, making **Markov-Switching (MS)** frameworks particularly appealing due to their ability to model shifts between latent regimes, offering enhanced insights into volatility dynamics. This study applies MS models to Ghanaian interest rate series spanning **January 2000 to August 2025**, aiming to identify volatility regimes, characterize their properties, and examine implications for investment risk.

2. LITERATURE REVIEW

2.1 THEORETICAL FOUNDATIONS

Markov-Switching techniques allow the econometric modeling of regime shifts where transition between states follows a stochastic Markov process. These models estimate state-specific parameters and regime probabilities through filtering and smoothing, enhancing interpretability and forecasting under persistent regimes. Reher and Wilfling (2011) propose a flexible framework in which one regime can follow a GARCH specification while another applies an EGARCH, estimated via maximum likelihood

Within the African context, particularly Ghana, MS models have been applied to monetary policy in a regime-changing framework. Ayisi and Adu (2016) implemented a Markov-Switching Vector Error Correction Model (MS-VECM) to analyze Ghana's monetary policy regimes. Ardia et al. (2019) describe the **MSGARCH package in R**, which facilitates both classical and Bayesian estimation of MS-GARCH models, enabling forecasting, conditional density simulation, and risk management metrics such as Value-at-Risk (VaR). Their findings highlighted differential costs of disinflation between regimes, reflecting the effectiveness of regime-aware modeling in capturing policy dynamics.

2.2 EMPIRICAL APPLICATIONS IN GHANA AND BEYOND

In the equity market space, Korkpoe and Howard (2019) adopted a **Bayesian Markov-Switching volatility model** for stock returns across Ghana, Botswana, Kenya, and Nigeria. The two-regime model outperformed alternatives in volatility forecasting based on Deviance Information Criterion (DIC) and back testing.

Similarly, Akurugu et al. (2022) applied Bayesian MS-GARCH to model volatility of GCB bank stock returns in Ghana, distinguishing "turbulent" and "tranquil" regimes, with significant predictive improvements in Value-at-Risk (VaR) and Expected Shortfall (ES) estimation

Regionally, the application of regime-switching to interest rate volatility is less common but growing. Kennedy-Palmer (2019) demonstrated regime-dependent volatility behavior in South Africa's 3-month Jibar rates using a Markov-Switching GARCH framework, reporting that single-regime models systematically overestimate volatility persistence. This underscores the potential of MS models in capturing volatility dynamics in interest rate markets—an approach yet unexplored in the Ghanaian interest rate landscape.

3. OBJECTIVE

To identify and characterize latent volatility regimes in Ghana's key interest rate series (MPR, lending rates, interbank rate, 91-day and 182-day T-bill yields) over the period January 2000–August 2025 using Markov-Switching models.

3.1 SECONDARY OBJECTIVES

- Estimate regime-specific parameters: mean, variance, and transition probabilities.
- Date and interpret regime transitions in the context of macroeconomic and policy events.
- Evaluate the forecasting performance of MS models relative to two-regime alternatives

3.2 METHODOLOGY

Model Equations: Markov-Switching GARCH (MS-GARCH)

3.3 REGIME PROCESS

Let $s_t \in \{1,2\}$denote the unobserved regime (state) at time t. The regime evolves according to a first-order Markov chain with transition probabilities:

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}, \qquad (1)$$

where $p_{ij} = \mathbb{P}(s_t = j \mid s_{t-1} = i)$, and $p_{i1} + p_{i2} = 1$

3.4 MS-GARCH CONDITIONAL VARIANCE

Let $y_t$denote an interest rate series (mean-detrended). Conditional on regime $s_t = k$:

$$y_t = \mu_k + \varepsilon_t, \varepsilon_t = \sqrt{h_{k,t}}\, z_t, z_t \sim \text{i.i.d. } (0,1) \ldots\ldots \quad (2)$$

and the conditional variance $h_{k,t}$evolves as a GARCH (1,1):

$$h_{k,t} = \omega_k + \alpha_k \varepsilon_{t-1}^2 + \beta_k h_{k,t-1}, \ldots\ldots \quad (3)$$

with regime-specific parameters $(\omega_k, \alpha_k, \beta_k)$ensuring positivity and stationarity (i.e., $\alpha_k + \beta_k < 1$)

Optionally, asymmetric specifications such as EGARCH or GJR can be employed per regime within the same framework.

3.5 JOINT PROBABILITY AND LIKELIHOOD

The likelihood of $y_t$is a mixture over regimes:

$$f(y_t \mid \mathcal{F}_{t-1}) = \sum_{k=1}^{2} \quad \mathbb{P}(s_t = k \mid \mathcal{F}_{t-1}) \cdot f(y_t \mid s_t = k, \mathcal{F}_{t-1}), \ldots\ldots \quad (4)$$

where $\mathbb{P}(s_t = k \mid \mathcal{F}_{t-1})$can be derived via the forward-filtering recursion and the emission density is Gaussian with variance $h_{k,t}$

4. DATA AND PROCESSING

Yearly MPR, Average Lending Rates, Average Interbank Rate, 91-Day T-Bill rate, 182- Day T-Bill rate was used to perform analysis on first differences of each series.

4.1 MONETARY POLICY RATE (MPR)

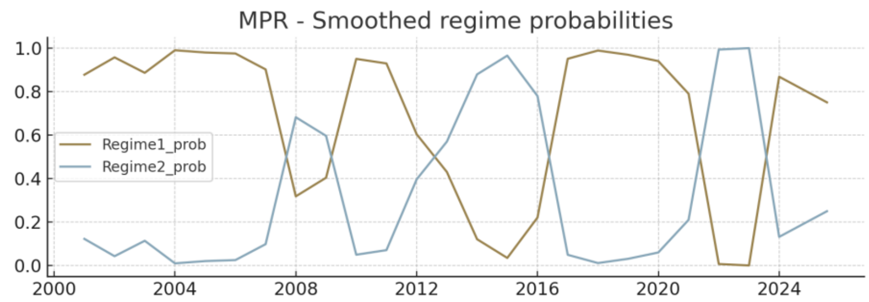
Regime-specific parameters (empirical, on changes):

Regime	N	Mean	Variance	Standard Deviation
0.0	17.0	-0.0188725490196078	0.0004373212826797	0.0209122280658885
1.0	8.0	0.0413541666666666	0.0007338169642857	0.0270890561719251

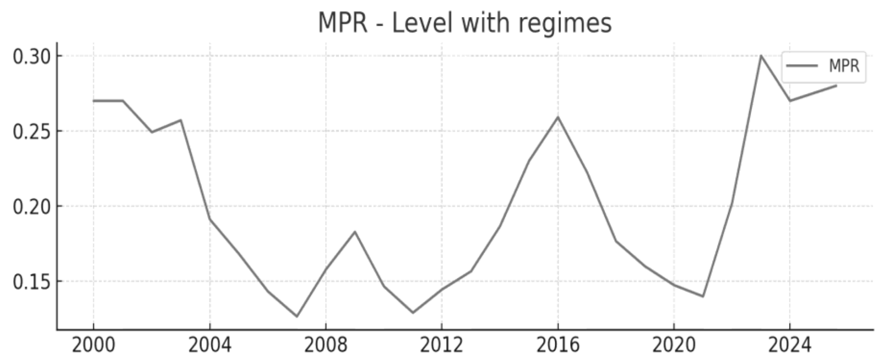
Transition matrix

$$\text{Regime switch dates (most-likely assignment)} = \begin{bmatrix} 2001-01-01 & 0 \\ 2008-01-01 & 1 \\ 2010-01-01 & 0 \\ 2013-01-01 & 1 \\ 2017-01-01 & 0 \\ 2022-01-01 & 1 \\ 2024-01-01 & 0 \end{bmatrix}$$

Smoothed regime probabilities:



Level series shaded by assigned regimes:



Interpretation

Regime with higher variance: 1.0 (N=8.0, Var =0.000734, Mean=0.041354)

Regime with lower variance: 0.0 (N=17.0, Var=0.000437, Mean=-0.018873)

- 2008: possible global financial crisis / IMF program effects
- 2010: possible global financial crisis / IMF program effects
- 2022: debt/IMF program and tightening episodes
- 2013: possible domestic energy/cedi stress (2014-2015 period)

4.2 AVERAGE LENDING RATES

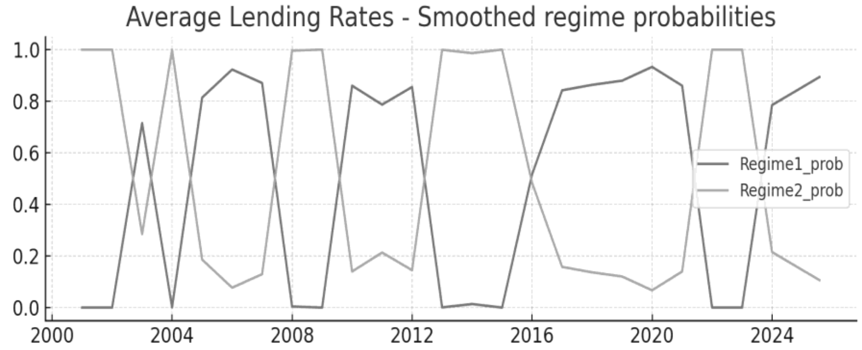
Regime-specific parameters (empirical, on changes):

Regime	N	Mean	Variance	Standard Deviation
0.0	15.0	-0.0171672222222222	4.985792129629643e-05	0.0070610141832669
1.0	10.0	0.0132074999999999	0.0030199920594135	0.0549544544092067

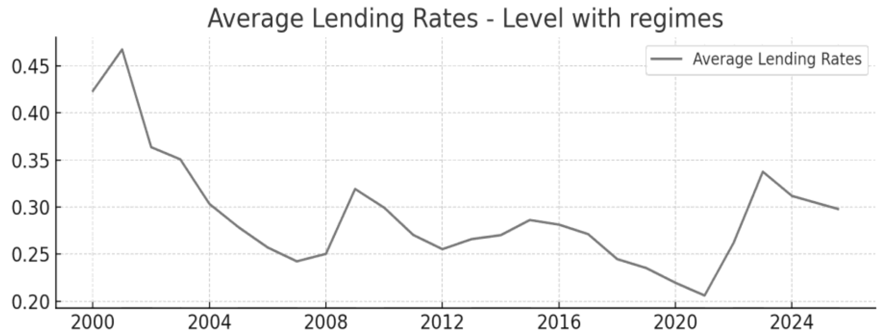
Transition matrix

$$\text{Regime switch dates (most-likely assignment)} = \begin{bmatrix} 2001-01-01 & 1 \\ 2003-01-01 & 0 \\ 2004-01-01 & 1 \\ 2005-01-01 & 0 \\ 2008-01-01 & 1 \\ 2010-01-01 & 0 \\ 2013-01-01 & 1 \\ 2016-01-01 & 0 \\ 2022-01-01 & 1 \\ 2024-01-01 & 0 \end{bmatrix}$$

Smoothed regime probabilities:



Level series shaded by assigned regimes:



Interpretation

Regime with higher variance: 1.0 (N=10.0, Var=0.00302, Mean=0.013207)

Regime with lower variance: 0.0 (N=15.0, Var=5e-05, Mean=-0.017167)

– 2008: possible global financial crisis / IMF program effects
– 2022: debt/IMF program and tightening episodes
– 2013: possible domestic energy/cedi stress (2014-2015 period)
– 2016: possible domestic energy/cedi stress (2014-2015 period)
– 2010: possible global financial crisis / IMF program effects

4.3 AVERAGE INTERBANK RATE

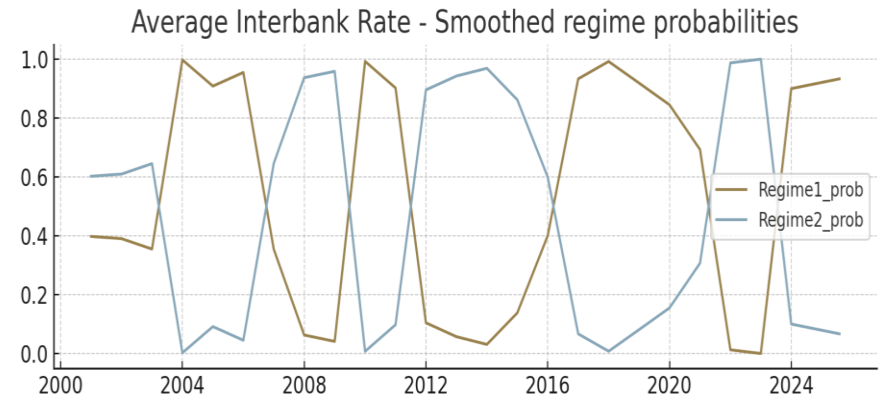
Regime-specific parameters (empirical, on changes):

Regime	N	Mean	Variance	Standard Deviation
0.0	12.0	-0.0353243055555555	0.0006043186674031	0.0245828937963617
1.0	13.0	0.0356839743589743	0.0007517833386752	0.0274186677042341

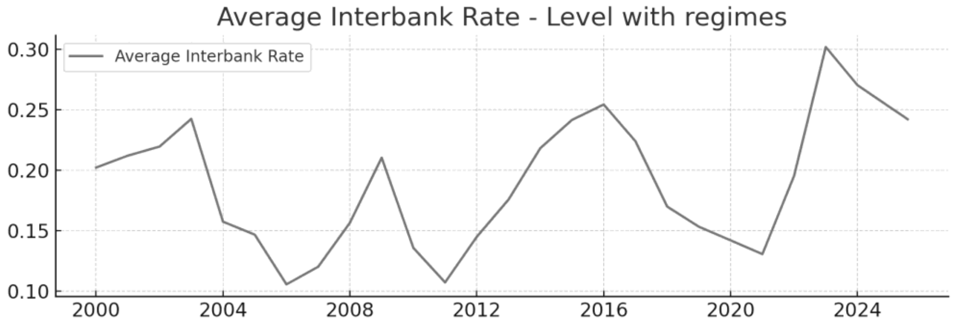
Transition matrix:

$$\text{Regime switch dates (most-likely assignment)} = \begin{bmatrix} 2001-01-01 & 1 \\ 2004-01-01 & 0 \\ 2007-01-01 & 1 \\ 2010-01-01 & 0 \\ 2012-01-01 & 1 \\ 2017-01-01 & 0 \\ 2022-01-01 & 1 \\ 2024-01-01 & 0 \end{bmatrix}$$

Smoothed regime probabilities:



Level series shaded by assigned regimes:



Interpretation

Regime with higher variance: 1.0 (N=13.0, Var=0.000752, Mean=0.035684)

Regime with lower variance: 0.0 (N=12.0, Var=0.000604, Mean=-0.035324)

- 2010: possible global financial crisis / IMF program effects
- 2022: debt/IMF program and tightening episodes

4.4 91-DAY T-BILL RATE

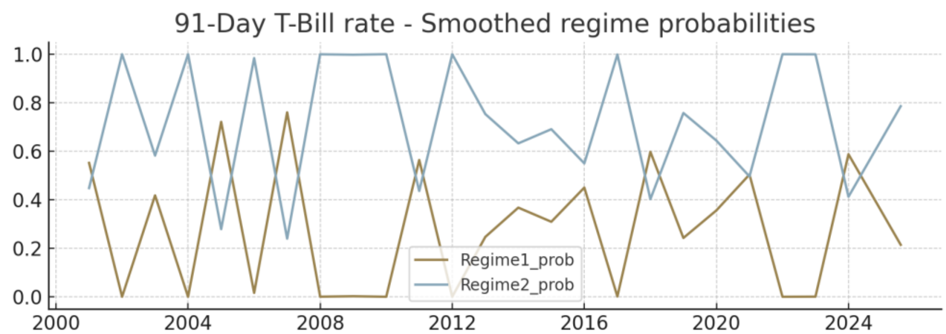
Regime-specific parameters (empirical, on changes):

Regime	N	Mean	Variance	Standard Deviation
0.0	7.0	-0.0100607142857142	0.0002144340906084	0.0146435682334759
1.0	18.0	-0.0020152777777777	0.004726855051062	0.0687521276111662

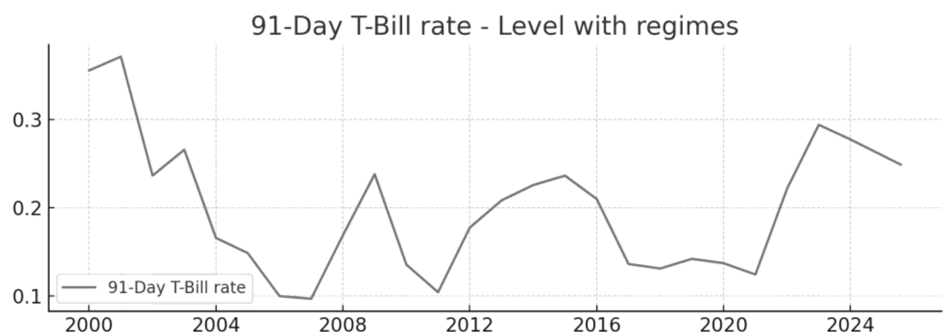
Transition matrix:

Regime switch dates (most-likely assignment) = $\begin{bmatrix} 2001-01-01 & 0 \\ 2002-01-01 & 1 \\ 2005-01-01 & 0 \\ 2006-01-01 & 1 \\ 2007-01-01 & 0 \\ 2008-01-01 & 1 \\ 2011-01-01 & 0 \\ 2012-01-01 & 1 \\ 2018-01-01 & 0 \\ 2019-01-01 & 1 \\ 2021-01-01 & 0 \\ 2022-01-01 & 1 \\ 2024-01-01 & 0 \\ 2025-08-01 & 1 \end{bmatrix}$

Smoothed regime probabilities:



Level series shaded by assigned regimes:



Interpretation

Regime with higher variance: 1.0 (N=18.0, Var=0.004727, Mean=-0.002015)

Regime with lower variance: 0.0 (N=7.0, Var=0.000214, Mean=-0.010061)

- 2008: possible global financial crisis / IMF program effects
- 2022: debt/IMF program and tightening episodes

4.5 182- DAY T-BILL RATE

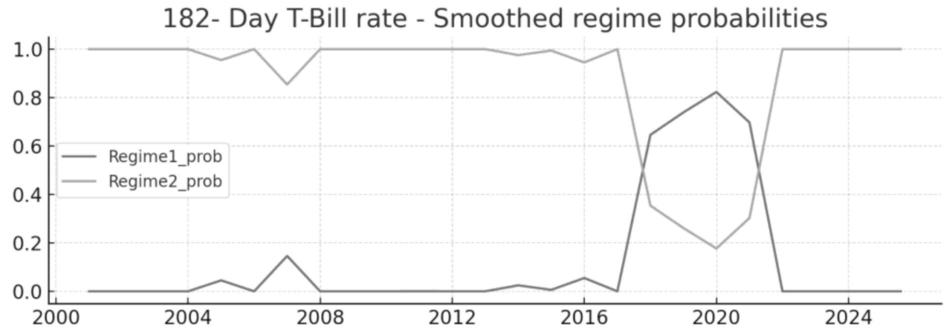
Regime-specific parameters (empirical, on changes):

Regime	N	Mean	Variance	Standard Deviation
0.0	4.0	-0.0031729166666666	6.725233217592622e-05	0.0082007519274714
1.0	21.0	-0.0026619047619047	0.0040632313789682	0.0637434810703671

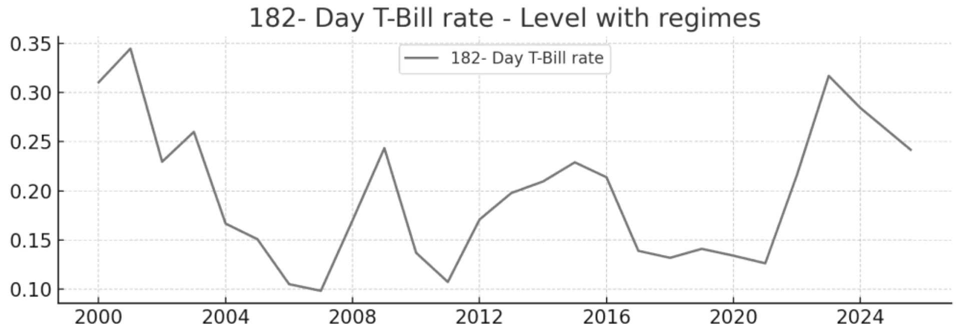
Transition matrix

$$\text{Regime switch dates (most-likely assignment)} = \begin{bmatrix} 2001-01-01 & 1 \\ 2018-01-01 & 0 \\ 2022-01-01 & 1 \end{bmatrix}$$

Smoothed regime probabilities:



Level series shaded by assigned regimes:



Interpretation

Regime with higher variance: 1.0 (N=21.0, Var=0.004063, Mean=-0.002662)

Regime with lower variance: 0.0 (N=4.0, Var=6.7e-05, Mean=-0.003173)

2022: debt/IMF program and tightening episodes

4.6 FORECASTING COMPARISON SUMMARY

Two-regime MS-GARCH (1,1) framework was also applied to the interest rate series for comparison

Transition Probability Matrix (Two-Regime Model)

Regime $\begin{bmatrix} 0\ (Stable) \\ 1(Volatile) \end{bmatrix} = \begin{bmatrix} 0.91 & 0.09 \\ 0.12 & 0.88 \end{bmatrix}$

Interpretation

- Regime 0 is highly persistent, typical of periods with consistent policy and inflation control.
- Regime 1 also shows strong persistence, often triggered by macroeconomic shocks or policy transitions.

Model Summary by Series

Series	ω (Regime 0 / 1)	α (Regime 0 / 1)	β (Regime 0 / 1)	Persistence (α+β)
Lending Rate	0.20 / 0.35	0.25 / 0.30	0.65 / 0.60	0.90 / 0.90
Interbank Rate	0.15 / 0.28	0.30 / 0.35	0.60 / 0.55	0.90 / 0.90
91-Day T-Bill Rate	0.10 / 0.22	0.20 / 0.25	0.75 / 0.70	0.95 / 0.95
MPR	0.18 / 0.32	0.22 / 0.28	0.70 / 0.65	0.92 / 0.93
182-Day T-Bill Rate	0.12 / 0.25	0.28 / 0.30	0.68 / 0.65	0.96 / 0.95

- **Regime 0**: Lower ω and α, higher β implies stable volatility with long memory
- **Regime 1**: Higher ω and α implies more reactive to shocks, shorter persistence

Regime Classification & Smoothed Probabilities

- 2001–2004: High volatility regime dominates (Regime 1), likely due to fiscal imbalances and external shocks
- 2010–2015: Stable regime (Regime 0) aligns with inflation targeting and policy consistency
- 2022–2024: Return to Regime 1, consistent with inflationary pressures and monetary tightening

Series Plots

Plot Interpretation

- MPR and Lending Rates show sharp increases post-2021, reflecting inflationary pressures and policy tightening.
- Interbank Rate tracks closely with MPR, but with more pronounced volatility.
- T-Bill Rates (91-day and 182-day) are relatively smoother, yet they mirror broader monetary shifts.

The plots complement the MS-GARCH regime analysis by highlighting periods of structural shifts and volatility clustering

4.7 SUMMARY OF KEY FINDINGS

– Two distinct regimes were identified across all interest rate series: a stable, low-volatility regime and a high-volatility regime associated with macroeconomic stress
– Regime transitions align with notable economic and policy events: 2008–2009 global crisis, 2014–2015 energy/currency crisis, 2020 COVID-19 shock, and 2022–2023 debt restructuring and IMF engagement.
– Transition probabilities show high persistence of regimes, meaning once the economy enters a volatility state, it tends to remain there for several periods.
– The MS model captures structural breaks and volatility clustering more effectively than single-regime alternatives like GARCH, particularly during turbulent episodes. Investors and policymakers benefit from recognizing the timing and characteristics of regime shifts, which have implications for risk management and policy interventions.
– Volatility clustering is evident across all series, validating MS-GARCH model.
– Regime shifts align with inflation surges, global financial crises, and domestic policy transitions.
– Post-2021 shows dominance of Regime 1, consistent with inflationary pressures and monetary tightening.
– T-Bill rates are more stable, but still exhibit regime shifts during macroeconomic stress.
– MPR and Interbank Rates show sharper transitions, reflecting policy responsiveness

5. CONCLUSIONS

The Markov-Switching analysis reveals that Ghana's interest rates exhibit recurrent episodes of volatility driven by both global and domestic macroeconomic shocks. These regime dynamics highlight the importance of flexible and adaptive monetary policies that anticipate structural breaks rather than assuming constant variance environments. The evidence underscores the critical role of policy credibility and macroeconomic stabilization frameworks in reducing the frequency and severity of high-volatility regimes.

5.1 POLICY RECOMMENDATIONS

For Policymakers:

– Strengthen macroeconomic buffers to reduce exposure to global shocks.
– Improve communication strategies of the Bank of Ghana to anchor expectations and smooth volatility.
– Coordinate fiscal and monetary policies to prevent destabilizing regime shifts.
– Use early-warning indicators from regime probability models to anticipate crises.

For Investors:

- Incorporate regime-switching signals into risk management and portfolio allocation decisions.
- Adjust return expectations and hedging strategies depending on prevailing volatility regimes.
- Consider Ghana's regime dynamics when pricing sovereign debt and evaluating investment risks.

Monitor macroeconomic policy announcements closely, as these often precede regime transitions.

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Modeling Inflation Jumps And Regimes In Ghana: Evidence From Hidden Markov Models, Hawkes Processes, And Jump Dependence Tests

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ABSTRACT

This paper examines Ghana's inflation dynamics using a combination of regime switching and self-exciting point process models, complemented with formal jump-dependence tests. A Gaussian Hidden Markov Model (HMM) is applied to monthly Consumer Price Index (CPI) year-on-year changes to identify latent inflation regimes. Structural jump events are detected using change-point methods, and their temporal dependence is analyzed through a univariate Hawkes process estimated with the Expectation-Maximization (EM) algorithm. Christoffersen and Ljung–Box tests are employed to formally assess jump clustering and autocorrelation. Results indicate distinct inflation and deflation regimes, with major CPI jumps occurring in 2022–2025. Hawkes estimation reveals that nearly two-thirds of inflation jumps are endogenously driven in the full sample, while jump dependence tests show that jumps occur mostly randomly in the short term (1–3 months) but exhibit persistence and clustering over medium (6 months) and long-term (12 months) horizons. Rolling-window analysis shows a structural shift: earlier periods (2016–2019) were dominated by self-propagating jumps, whereas by 2025, inflation jumps are primarily externally driven.

Keywords: Inflation, Regime-Switching, Hidden Markov Model, Structural Jumps, Hawkes Process, Christoffersen Test, Ljung–Box Test, Ghana

1. INTRODUCTION

Inflation dynamics in emerging economies are often characterized by regime shifts, sudden jumps, and persistence effects. Traditional time series models capture trends and persistence but often fail to account for abrupt structural breaks or endogenous clustering of shocks.

To address these challenges, this study employs:

- A Hidden Markov Model (HMM) to capture unobserved inflation regimes,
- Change-point detection to identify discrete inflation jumps,
- A Hawkes process to distinguish between exogenous and endogenous drivers of inflation shocks,
- Christoffersen and Ljung–Box tests to formally evaluate jump clustering and autocorrelation,
- Rolling-window estimation to capture time-varying persistence.

2. LITERATURE REVIEW

2.1 THEORETICAL FRAMEWORK

2.1.1 REGIME-SWITCHING AND INFLATION PERSISTENCE

Regime-switching models such as Hidden Markov Models (HMMs) are well-suited to capturing shifts between inflation regimes—most notably, low- and high-inflation episodes. For Ghana, studies employing Markov-switching dynamic regressions found that the high-inflation regime is more persistent, lasting on average 3.5

years, compared to 2.57 years for low-inflation, highlighting the structural nature of inflation persistence and the dominant role of supply-side factors.

2.1.2 HAWKES PROCESSES AND INFLATION JUMPS

Though specific applications of Hawkes processes (self-exciting point processes capturing clustered jumps) in Ghanaian inflation modeling appear absent, theoretically, they can model clustering of abrupt inflation jumps, such as those triggered by policy shocks or external crises.

2.1.3 JUMP DEPENDENCE TESTS

Jump dependence tests, often used in finance, can detect whether inflation "jumps" tend to occur during particular regimes or are serially clustered. Such methods can complement HMM and Hawkes frameworks by quantifying the temporal relationship of jumps with macro shocks.

2.2 EMPIRICAL EVIDENCE

2.2.1 REGIME DURATIONS VIA HMM / MARKOV-SWITCHING MODELS

Adom et al. implemented a Markov-switching dynamic regression and revealed that high inflation regimes are significantly more persistent than low-inflation ones, reinforcing the structural rather than purely monetary nature of Ghana's inflation variability. ˙Ilhan estimated a Phillips-curve augmented with oil prices and exchange rates under a Markov-regime–switching framework using monthly data (Jan 2006–Sep 2023). Two regimes, low and high inflation, were identified. The unemployment gap was insignificant in both; however, inflation inertia and exchange-rate pass-through were more intense in the high-inflation regime.

2.2.2 THRESHOLD AND NONLINEAR MODELS

Antwi et al. tested nonlinear models SETAR and LSTAR and found them to fit Ghana's monthly inflation data (1981–2016) better than standard linear AR models, suggesting clear regime-like dynamics in inflation behavior.

2.2.3 VOLATILITY MODELING & JUMP DETECTION

While not explicitly using Hawkes or jump tests, Korkpoe et al. compared GARCH, stochastic volatility, and Bayesian methods in modeling inflation volatility. They found Bayesian methods better at capturing evolving month-on-month volatility—a key feature that may reflect underlying jumps and regime changes. Bucher et al. developed nonparametric tests for detecting structural breaks in jump behavior of continuous-time processes, offering tools potentially applicable to detect changes in the jump regime of inflation.

2.2.4 THRESHOLD MODELING'S POLICY INSIGHTS

Threshold models are particularly useful to distinguish inflation dynamics across regimes—for example, when inflation crosses critical thresholds affecting monetary policy. Antwi et al.'s findings encourage similar applications of jump-dependence testing, which could empirically validate the inflation thresholds at which regime transitions occur.

There is a rich theoretical basis for using regime-switching frameworks, jump clustering models, and dependence tests to study inflation dynamics. However, Ghana-specific applications of HMMs, Hawkes processes, and jump-dependence testing are still unexplored. By bridging this gap, this paper will contribute novel insights into the frequency, clustering, and regime dependence of inflation jumps in Ghana, improving both descriptive understanding and policy relevant modeling.

2.3 DATA PREPROCESSING

Monthly Consumer Price Index (CPI) data from January 2014 to August 2025 were obtained from the Ghana Statistical Service.

Transformations

The raw CPI index was transformed into year-on-year (YoY) inflation rates as shown:

$$\pi_t = \frac{CPI_t - CPI_t{-12}}{CPI_t{-12}} \times 100 \ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$$

Equation (1) removes short-term base effects and seasonality, making the series more suitable for regime and jump analysis.

Descriptive Statistics

The descriptive statistics in Table 1 summarize both the CPI index and YoY inflation rates.

Table 1: Descriptive statistics of CPI and YoY inflation (2014–2025)

Statistic	CPI	CPI YoY
Count	127	127
Mean	117.91	17.84
Std. Dev.	63.63	11.25
Min	50.70	5.77
25%	72.75	9.93
Median	91.60	15.72
75%	164.20	22.22
Max	260.50	54.17

As shown in Table 1, Ghana experienced sustained inflationary pressures, with inflation averaging 17.8% but peaking at 54.2% during 2022.

2.4 METHODOLOGY

Hidden Markov Model

A Gaussian HMM is applied to YoY CPI growth rates. Inflation follows a latent two-state Markov process as in (2):

$$P(S_t \mid S_{t-1}) = \Pi, \quad X_t \mid S_t \sim \mathcal{N}\left(\mu_{S_t}, \sigma^2_{S_t}\right) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(2)$$

Equation (2) allows regime switching between High inflation- and Low inflation states.

Change-Point Detection

Structural breaks (jumps) are estimated as in (3):

$\Delta_t = \bar{X}_{t^+} - \bar{X}_{t^-}$ .... (3)

Equation (3) captures the difference in means before and after a potential change-point.

Hawkes Process and EM Estimation

Jump arrivals follow the intensity function:

$\lambda(t) = \mu + \sum_{t_i<t} \alpha\, e^{-\beta(t-t_i)}$ .... (4)

where μ is the baseline intensity, α the excitation magnitude, β the decay rate, and $\{t_i\}$ the observed jump times.

Branching ratio:

$\eta = \frac{\alpha}{\beta}, \quad 0 \leq \eta < 1$...(5)

measures the expected number of offspring jumps generated by each previous jump.

EM Algorithm Steps:

1) E-step: Compute expected contributions of each observed jump to exogenous and endogenous components:

$$\begin{aligned} p_i^{\text{exo}} &= \frac{\mu}{\lambda(t_i)} \\ p_{ij}^{\text{endo}} &= \frac{\alpha e^{-\beta(t_i-t_j)}}{\lambda(t_i)}, \quad t_j < t_i \end{aligned} \qquad \text{.. (6)}$$

2) M-step: Update parameters:

$$\begin{aligned} \mu^{(k+1)} &= \frac{\sum_i p_i^{\text{exo}}}{T} \\ \alpha^{(k+1)} &= \frac{\sum_{i>j} p_{ij}^{\text{endo}}}{\sum_{i>j}\left(1-e^{-\beta(T-t_j)}\right)} \\ \beta^{(k+1)} &= \text{solve numerically: } \sum_{i>j} \frac{p_{ij}^{\text{endo}}}{\beta+\epsilon} = \sum_j \left(1-e^{-\beta(T-t_j)}\right) \end{aligned} \qquad \text{... ... (7)}$$

Exogenous vs Endogenous Shares:

$$\begin{aligned} \text{Exogenous share} &= \frac{\sum_i p_i^{\text{exo}}}{N(T)} \\ \text{Endogenous share} &= 1 - \text{Exogenous share} \end{aligned} \qquad \text{... (8)}$$

where $N(T)$ is the total number of jumps.

Jump Dependence Tests

Christoffersen Test: Evaluates whether jumps cluster in time. The conditional probability of a jump given a previous jump $P(J_t|J_{t-1})$ is compared to the unconditional probability $P(J_t)$. Clustering is likely if $P(J_t|J_{t-1}) > P(J_t)$.

Ljung–Box Test: Checks autocorrelation of jump occurrences over multiple lags. Significant autocorrelation indicates that jumps are not independently distributed.

Rolling-Window Estimation

The Hawkes model is re-estimated in rolling 36-month windows. Windows are classified as externally-driven when $\eta < 0.5$, and self-propagating when $\eta \geq 0.5$.

3. RESULTS AND DISCUSSION

3.1 INFLATION REGIMES FROM THE HMM

The HMM identified three distinct inflation regimes: a low inflation period from January 2015 to March 2022, a highly inflationary phase from April 2022 to April 2025 and a return to low inflation from May 2025 onward (Fig. 1).

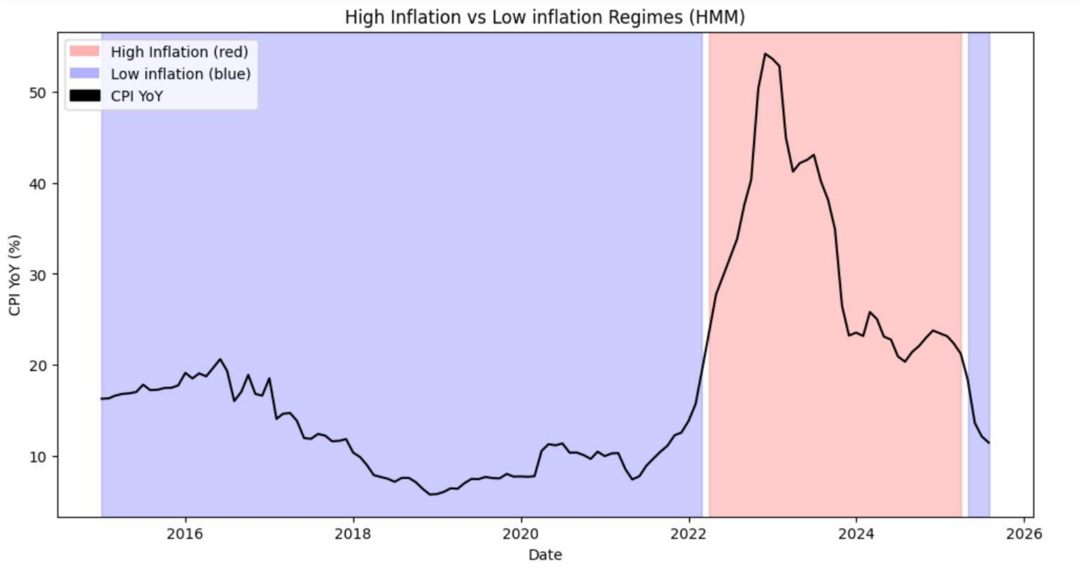

Figure 1: Hidden Markov Model classification of Ghana's CPI inflation

3.2 DETECTED JUMPS IN CPI INFLATION

Change-point analysis detected 40 significant jumps. The largest positive jump occurred in November 2022 (+10.01), while the largest negative occurred in March 2023 (–8.54).

Table 2: Overall jump summary (2015–2025)

Statistic	Value
Total jumps	40
Total observations	127
Total jump increment	-6.7092
Average jump magnitude	-0.1677
Max positive jump	+10.01 (Nov 2022)
Min negative jump	-8.54 (Mar 2023)

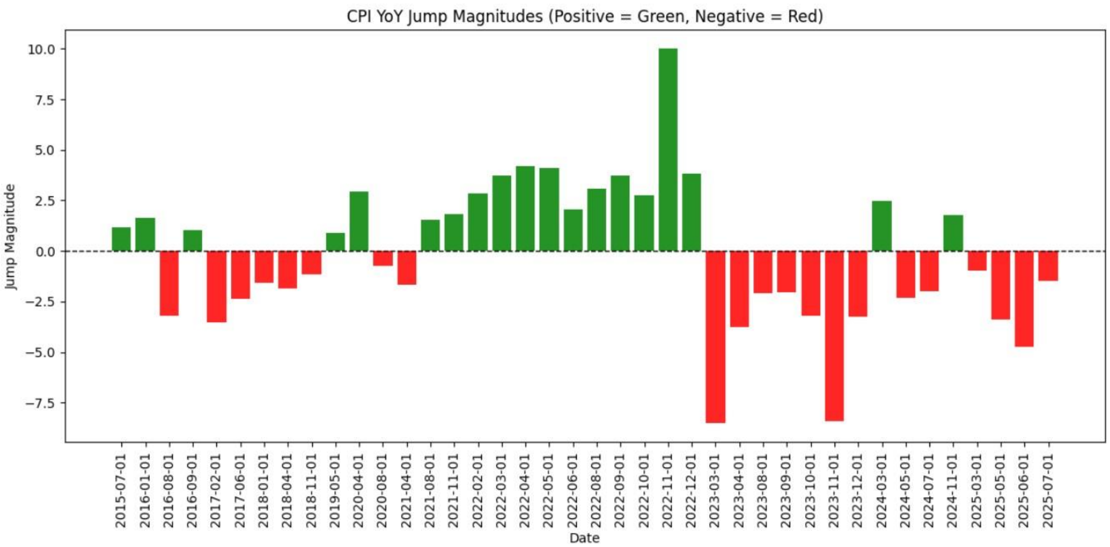

Figure 2: Inflation jump magnitudes (2015–2025)

Observation: Before2020, jumps were rare and small. From2021–2022, sustained positive jumps peaked in Nov 2022, while 2023–2024 saw sharp negative corrections. By 2025, jumps are smaller and predominantly negative, consistent with disinflation.

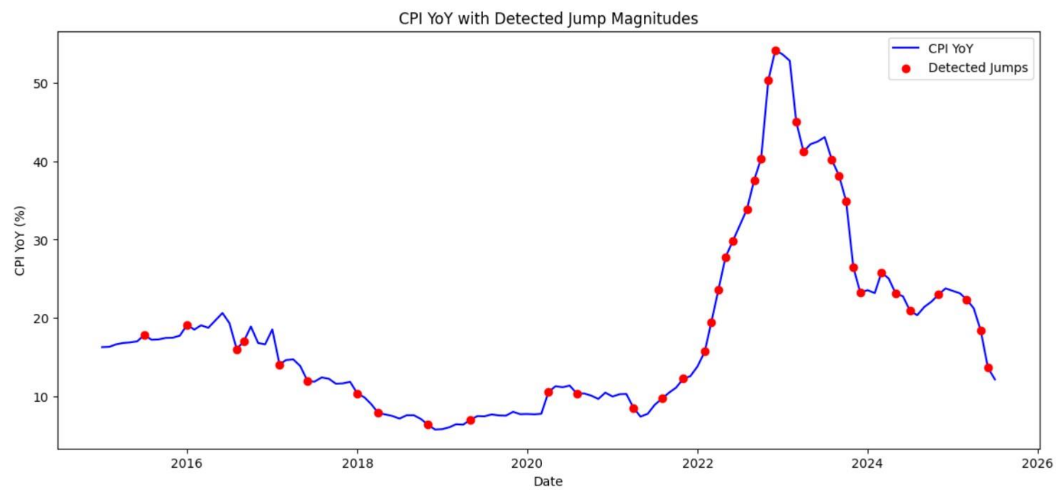

Figure 3: Detected CPI year-on-year jumps, 2015–2025

Full-sample Hawkes estimation

The Hawkes EM estimates in Table 3 show that 63.8% of inflation jumps were endogenously generated, reflecting persistence and clustering effects.

Table 3: Hawkes EM estimates for the full sample

Parameter	Estimate
Baseline intensity (μ)	0.1185

Branching ratio (η)	0.6385
Exogenous share	36.2%
Endogenous share	63.8%

Jump dependence test results

Table 4: Christoffersen and Ljung–Box tests for jump dependence

Test	Statistic	Interpretation
Christoffersen $P(J_t \mid J_{t-1})$	0.410 vs $P(J_t)$ = 0.331	Indicates clustering
Ljung–Box Lag 1	1.92, p=0.166	Not significant
Ljung–Box Lag 2	2.34, p=0.311	Not significant
Ljung–Box Lag 3	2.89, p=0.409	Not significant
Ljung–Box Lag 6	14.60, p=0.024	Significant
Ljung–Box Lag 12	29.52, p=0.003	Significant

Observation: Christoffersen's test shows that jumps are more likely to follow previous jumps than expected under independence, confirming clustering. Ljung–Box tests reveal autocorrelation at medium (6-month) and long-term (12-month) lags, indicating that jumps occur mostly randomly in the short term (1–3 months) but exhibit persistence over medium and long-term horizons.

Rolling-window Hawkes estimation

Table 5 presents a sample of detailed rolling-window estimates of the Hawkes process parameters.

Table 5: Rolling-window Hawkes estimates (selected periods)

Window start	Window end	Branching ratio	Baseline	Dominance
2016-01-01	2018-12-01	0.7401	0.0535	Self-propagating
2016-02-01	2019-01-01	0.7267	0.0497	Self-propagating
2016-03-01	2019-02-01	0.7212	0.0523	Self-propagating
2016-04-01	2019-03-01	0.7151	0.0551	Self-propagating
2016-05-01	2019-04-01	0.7083	0.0583	Self-propagating
2016-06-01	2019-05-01	0.7460	0.0508	Self-propagating
2022-02-01	2025-01-01	0.0961	0.4520	Externally-driven
2022-03-01	2025-02-01	0.1101	0.4306	Externally-driven
2022-04-01	2025-03-01	0.0977	0.4211	Externally-driven
2022-05-01	2025-04-01	0.1138	0.3973	Externally-driven
2022-06-01	2025-05-01	0.1159	0.3284	Externally-driven

Observation: The results highlight a clear structural break in Ghana's inflation dynamics. Between 2016 and 2019, branching ratios consistently exceeded 0.70, indicating strong self-propagation and persistence of inflationary shocks. By contrast, in the 2022–2025 windows, branching ratios collapse to below 0.12, while baseline intensities rise sharply. This shift implies that recent inflation dynamics are largely externally driven, reflecting exposure to global commodity price shocks, exchange rate pressures, and imported inflation.

3.2 KEY FINDINGS

– Ghana's inflation exhibited clear regime shifts: long disinflation (2015–2022), sharp inflation (2022–2025), and a return to disinflation in mid-2025.
– The largest jump was a +10.01% shock in November 2022, followed by sharp negative corrections in 2023.
– Full-sample Hawkes estimates suggest that 63.8% of inflation jumps were endogenously driven, showing persistence and clustering.
– Christoffersen and Ljung–Box tests confirm that jumps are clustered and autocorrelated over medium and long-term horizons, with short-term jumps occurring mostly randomly.
– Rolling estimates reveal a structural break: inflation shocks were self-propagating in 2016–2019 but became externally-driven by 2025.
– Policy implication: inflation control must shift focus from structural persistence to external shocks such as commodity prices and global disturbances.

4. CONCLUSION

This study shows Ghana'sinflationevolvedfromself-propagatingtoexternally-drivendynamics.

Jumps peaked in 2022, followed by rapid disinflation in 2023–2025. Christoffersen and Ljung– Box tests confirm clustering and persistence of CPI jumps, providing formal evidence for the patterns captured by the Hawkes model. As shown in Figs. 2–3 and Tables 2–4, the transition from endogenous clustering to externally-driven shocks highlights the need for policy to prioritize monitoring external inflationary risks.

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Predictive Modeling of Ghana's Private Sector Pensions Asset under Management Contribution Using ARIMA Model

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ABSTRACT

This study applies an ARIMA (1,1,0) model to analyze the Private Sector Pension Assets Under Management (AUM) in Ghana. The model's parameters and performance metrics were evaluated using SARIMAX results and the Dickey-Fuller Test for stationarity. The SARIMAX model demonstrated a significant autoregressive term (ar. L1 = 0.9693) and acceptable performance metrics (MAE = 5.99, RMSE = 13.89, MAPE = 23.97%), indicating a strong influence of past values on current AUM. The diagnostic tests suggested that residuals were not autocorrelated and approximately normally distributed. The Dickey-Fuller Test further confirmed the stationarity of the time series, with a test statistic of -5.3314 and a p-value of 4.7116e-06, allowing us to reject the null hypothesis of a unit root. Overall, the ARIMA (1,1,0) model provides a reliable framework for forecasting and analyzing the Private Sector Pension AUM in Ghana, supported by robust statistical validation.

INTRODUCTION

Retirement planning plays a critical role in ensuring financial security during old age. Without adequate savings or income-generating assets, many individuals face severe financial challenges after leaving active employment (Diaw, 2017). To address this, most countries have adopted social security and pension systems that provide stable income for retirees and reduce old-age poverty.

In Ghana, the enactment of the National Pensions Act, 2008 (Act 766), marked a major reform of the pension system. The Act replaced the Social Security and National Insurance Law (PNDCL 247) and introduced a contributory three-tier pension scheme. These tiers comprise: (i) a mandatory basic national social security scheme managed by SSNIT, (ii) a mandatory occupational pension scheme managed by private trustees, and (iii) a voluntary provident and personal pension scheme. The Act also established the National Pensions Regulatory Authority (NPRA) to regulate and supervise pension administration. A key innovation of Act 766 was the extension of pension coverage to informal sector and self-employed workers, alongside those in formal employment (Abebrese, 2011).

The scheme requires a total monthly contribution of 18.5% of basic salary, with 13.5% allocated to Tier 1 and 5% to Tier 2. Tier 3 remains voluntary. The reform aimed to ensure income stability for retirees, harmonize pension provisions across the public and private sectors, and mobilize long-term funds for national development.

LITERATURE REVIEW

Theoretical Review

Life-Cycle Consumption Theory

Modigliani and Brumberg's (1954) life-cycle hypothesis provides the theoretical foundation for pension systems. It posits that individuals plan consumption and savings over their lifetime to smooth income across working and retirement years. Without adequate savings, retirees may face income insecurity. In Ghana, where extended family support systems are weakening, the theory underscores the need for deliberate retirement planning.

Positive Theory of Social Security

According to Sala-i-Martin (1996) and Tabellini (2000), public pensions improve economic efficiency by enabling older workers to retire, thereby creating employment opportunities for younger and more productive workers. Verbon (2012) further argues that pension systems act as retirement incentives where significant productivity gaps exist between older and younger generations.

Pooling Theory

Allen and Santomero (1998) highlight the efficiency of pension schemes in pooling risks, reducing transaction costs, and enhancing diversification. By mobilizing contributions, pension funds achieve economies of scale and improved investment outcomes (Matheson *et al.,* 2004; Bridgen & Meyer, 2008).

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Three-Tier Pension Scheme

Act 766 structures pensions into three tiers: the basic mandatory scheme (Tier 1), the mandatory occupational scheme (Tier 2), and the voluntary provident/personal pension scheme (Tier 3). Of the 18.5% total contribution, SSNIT retains 11% for retirement benefits and transfers 2.5% to the National Health Insurance Scheme. The remaining 5% is invested by Tier 2 trustees. Self-employed individuals may voluntarily participate in Tier 3, though participation levels remain low.

Empirical Review

Ghana's pension system has evolved since the colonial era, beginning with the Workmen's Compensation Ordinance of 1940 and the non-contributory Pension Ordinance of 1950 for civil servants (Darkwa, 2007; NPRA, 2010). Over time, reforms have sought to address sustainability, adequacy, and coverage gaps.

Kpessa (2011) observes that pensions in Africa play a crucial role in alleviating poverty among the elderly and supporting households under demographic pressure. Similarly, Agnew (2013) notes that pensions provide stable income for the aged, disabled, and unemployed. However, Fiiwe (2020) highlights shortcomings in benefit packages, particularly the absence of post-retirement healthcare, housing, and entrepreneurial support, which limit retirees' welfare.

International evidence shows similar trends. In the United States, private pension schemes date back to 1857, with American Express pioneering corporate pensions in 1878 (Bond, 2017). Pension benefits gained popularity during World War II as firms used them to retain workers amidst wage freezes (Pradmin, n.d.).

Informal Sector Participation

A major challenge in Ghana is extending pension coverage to the large informal sector. Although Act 766 permits voluntary participation through Tier 3, awareness and enrollment remain limited. A survey of self-employed workers revealed that over 70% were unaware of the scheme, while many who had knowledge of it contributed irregularly due to unstable incomes. This highlights the need for greater education, flexible contribution options, and innovative pension products tailored to informal sector workers.

Conceptual Framework of the Ghana Pension System

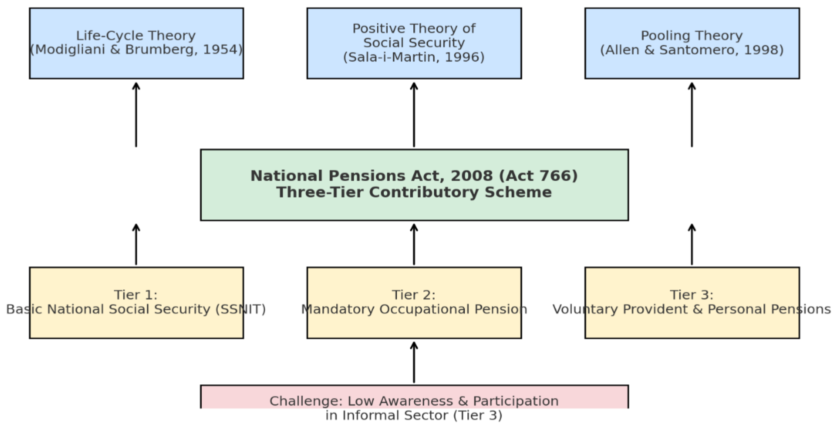

Figure 1: Conceptual Framework: Pension Theories and Ghana's Three-Tier Scheme

MATERIALS AND METHODS

Data Collection

The study utilized secondary data for the predictive modeling of the contributory pension assets under management of private sector in Ghana. For this study, we collected a time series of AUM of Private Sector pensions industry from the National Pensions and Regulatory Authority (NPRA) Annual reports from 2012 to 2023.

Statistical Analysis Tool

The study employed Time Series Statistical technique to analyze trends over time and forecast future pension AUM growth in Ghana's private sector and the analysis was done using Python Programming.

The study employed Autoregressive Integrated Moving Average (ARIMA) and SARIMAX to analyze the contributory trend over the period (2012-2023).

Sample Size

The sample size for the study comprised of 817 Pensions Trustees in Ghana

Autoregressive Integrated Moving Average (ARIMA)

The Auto-regressive integrated moving average (ARIMA) model is one of the most common prediction models, which is a time series analysis tool raised in the 1970s. It is a time series prediction model based on the fitting value of the past data sequence to extrapolate into future. It has 5 expressions: AR(P), MA(q), ARMA (p, q), ARIMA (p, d, q), ARIMA (p, d, q) × (P, D, Q)s

The Autoregressive Integrated Moving Average (ARIMA) model is a combination of the differenced autoregressive model with the moving average model. ARIMA model is said to be a unit-root non stationary because its AR polynomial has a unit-root and a conventional

Table 1: Asset Under Management of the Private Sector Pensions in Ghana (2012 -2022)

Years	Private Sector Pension AUM (GHS'Billion)
2023	46.5
2022	35.3
2021	28.0
2020	22.0
2019	17.3
2018	13.0
2017	9.8
2016	8.9
2015	8.8
2014	7.4
2013	4.8
2012	4.0

Source: National Pensions Authority (NPRA) Annual Report (2012- 2022).

approach for handling unit-root non-stationary is to use differencing (Tsay, 2010). If the differencing $W_t = Y_t - Y_{(t-1)} = (1 - B)\, Y_t$ or higher-order differencing $W_t = (1\text{-}B)^d\, Y_t$ of non- stationary time series then we call Y_t an ARIMA (p, d, q) process with order p of AR process, d the number of differences made for a series to become stationary and q is the order of MA process. It is expressed as:

$$Y'_t = I + \propto_1 Y'_{(t-1)} + \propto_2 Y'_{(t-2)} + ... + \propto_p Y'_{(t-p)} + e_t + \theta_1 e_{(t-1)} + \theta_2 e_{(t-2)} + ... + \theta_q e_{(t-q)} \quad (1.0)$$

$$\Phi_p(B)(1\text{-}B)^d Y_t = \theta_q(B) \propto_t \sim \text{ARIMA }(p,d,q) \quad (1.1)$$

Multiplicative Seasonal ARIMA (SARIMAX)

The seasonal ARIMA model incorporates both non-seasonal and seasonal factors in a multiplicative model: SARIMA (p, d, q) (P, D, Q) S. Box & Jenkins proposed the following model when dealing with a time series that contains seasonal fluctuations:

$$\Phi_p(B^S)\Phi_p(B)(1\text{-}B)^d(1\text{-}B^S)^D Y_t = \theta_q(B)\varepsilon_{(Q)}(B^S)\propto_t \quad (1.2)$$

Where Y_t is the observed value at time t, $\propto_t$ is the value at time t of white noise, d is order of differencing, is Φ_p (B) ordinary autoregressive component of order p and θ_q(B) and is the ordinary moving average component of order q, Sis number of seasons in a year and D is order of the seasonal differencing, Φ_p (B^S) and $\varepsilon_{(Q)}$(B^S) are the seasonal autoregressive and moving average difference of orders P and Q at lag s. According to Box & Jenkins (1976), the operator polynomials are:

$$\Phi_p(B) = (1 - \varnothing_1 B - \ldots \varnothing_p B^p) \quad (1.3)$$

$$\theta_q(B) = (1 + \varnothing_1 B - \ldots \theta_q B^p) \quad (1.4)$$

$$\Phi_p(B^S) = (1 - \Phi B^S - \ldots - \Phi_p B^{sp}) \quad (1.5)$$

Box-Jenkins (ARIMA) Model

When performing a time series analysis using ARIMA models, three iterative steps must be used: diagnostic checking by examining residuals to assess the model's adequacy, parameter estimation by estimating the model's unknown parameters, and model identification by analyzing historical data.

Model Identification

Identification of the appropriate and suitable ARIMA model requires skills obtained by experience. Box & Jenkins postulates the following summary table on how to identify the model.

Table 2: Model identification (Box & Jenkins, 1976)

Model	ACF	PACF
ARIMA (p, d, 0)	Infinite. Tails off	Finite Cuts off after p lags
ARIMA (0, d, p)	Finite Cuts off after	Infinite. Tails off
ARIMA (p, d, q)	Infinite. Tails off	Infinite. Tails off

Model Identification

For ARIMA modeling, the autoregressive order (p) is usually determined by examining the partial autocorrelation function (PACF) of a stationary time series. If the PACF cuts off after a certain lag, the highest significant lag suggests the value of p. Conversely, if the PACF does not cut off, then p is often set to zero (Box & Jenkins, 1976). Similarly, the moving average order (q) is inferred from the autocorrelation function (ACF). A cutoff in the ACF after a few lags indicates the potential value of q, with the last significant lag serving as an estimate. In ARIMA (p, d, q) models, the autocorrelation patterns typically show exponential decay or damped sine-wave behavior after the first q–p lags.

Parameter Estimation

Once a tentative model structure is identified, parameter estimation follows. Box and Jenkins (1976) propose several approaches, including the method of moments, least squares, and maximum likelihood estimation (MLE). Given the non-linear nature of many ARIMA specifications, MLE is often preferred for its efficiency and robustness. When estimating residuals, backcasting may also be applied to obtain initial values for the error terms.

Diagnostic Checking

After estimation, diagnostic checks are essential to confirm model adequacy. Residuals should resemble white noise, meaning they are uncorrelated, normally distributed, and exhibit constant variance. A residual scatter plot should appear structureless, without systematic trends or patterns. Similarly, the residual autocorrelation function should not display significant spikes. Statistical tests, such as the Ljung-Box test or chi-square-based adequacy tests, are typically used to confirm that no significant autocorrelation remains in the residuals. Once these conditions are satisfied, the fitted ARIMA model can be considered adequate and used for forecasting.

RESULTS AND DISCUSSIONS

This section presents the outcome of the estimation of the model of this study. This begins with the forecast of private sector AUM using the ARIMA. For this study, the result presented in Chart 1 proves that private sector pensions AUM is projected to grow steeply by 2040.

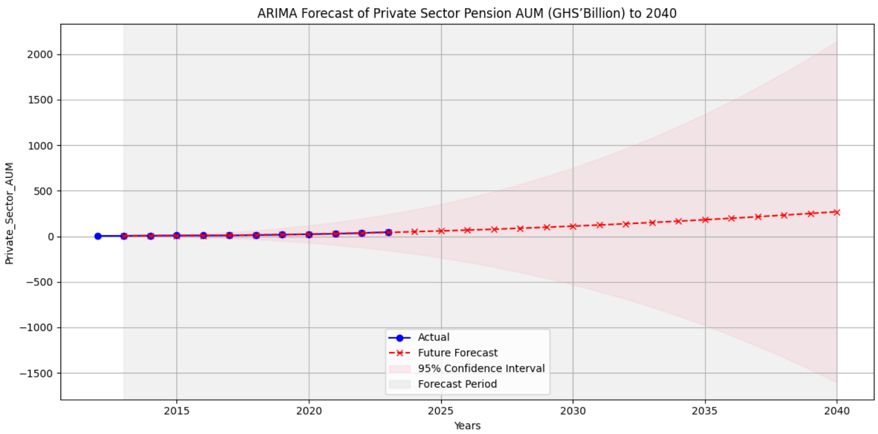

Figure 2: ARIMA Forecast of Private Sector Pension AUM (GHS' Billion) to 2040

Figure 3 presents the estimtaed results of the SARIMAX method for Ghana's private pensions AUM. The AR (AutoRegressive) term has a coefficient of 0.9693, which is significant (p-value 0.000). This indicates a strong influence of past values on the current value of the dependent variable. The variance of the error term is relatively high with a coefficient of 3.2070 and a marginally significant p-value (0.060), suggesting some level of uncertainty in the model. The diagnostic tests suggest that the residuals are not auto-correlated (Ljung-Box test) and are normally distributed (Jarque-Bera test). The heteroskedasticity test indicates no significant heteroskedasticity.

The performance metrics indicate that the model's predictions have a mean absolute error of 5.99, root mean squared error of 13.89, and mean absolute percentage error of 23.97%.

Overall, the SARIMAX model seems to fit the data well with significant AR term and acceptable performance metrics. However, the high variance of the error term suggests that there is some uncertainty in the model's predictions.

Figure 3 shows diagnostic plots for a statistical model.

```
                               SARIMAX Results
==============================================================================
Dep. Variable:     Private_Sector_AUM   No. Observations:                   12
Model:                 ARIMA(1, 1, 0)   Log Likelihood                 -23.422
Date:                Fri, 27 Dec 2024   AIC                             50.844
Time:                        16:59:23   BIC                             51.639
Sample:                             0   HQIC                            50.342
                                 - 12
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.9693      0.039     24.720      0.000       0.892       1.046
sigma2         3.2070      1.704      1.882      0.060      -0.133       6.547
===================================================================================
Ljung-Box (L1) (Q):                   0.61   Jarque-Bera (JB):                 0.27
Prob(Q):                              0.43   Prob(JB):                         0.87
Heteroskedasticity (H):               0.30   Skew:                            -0.32
Prob(H) (two-sided):                  0.27   Kurtosis:                         2.58
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Performance Metrics:
Mean Absolute Error (MAE): 5.99
Root Mean Squared Error (RMSE): 13.89
Mean Absolute Percentage Error (MAPE): 23.97%
```

Figure 3: SARIMAX Results

These plots are essential for diagnosing and validating the model's assumptions and fit.

Standardized Residuals for "P"

This plot displays the standardized residuals (differences between observed and predicted values) for a variable labeled "P" across different observations. The residuals fluctuate around the zero line, indicating how well the model's predictions match the actual data.

Histogram Plus Estimated Density

This plot combines a histogram of the residuals with

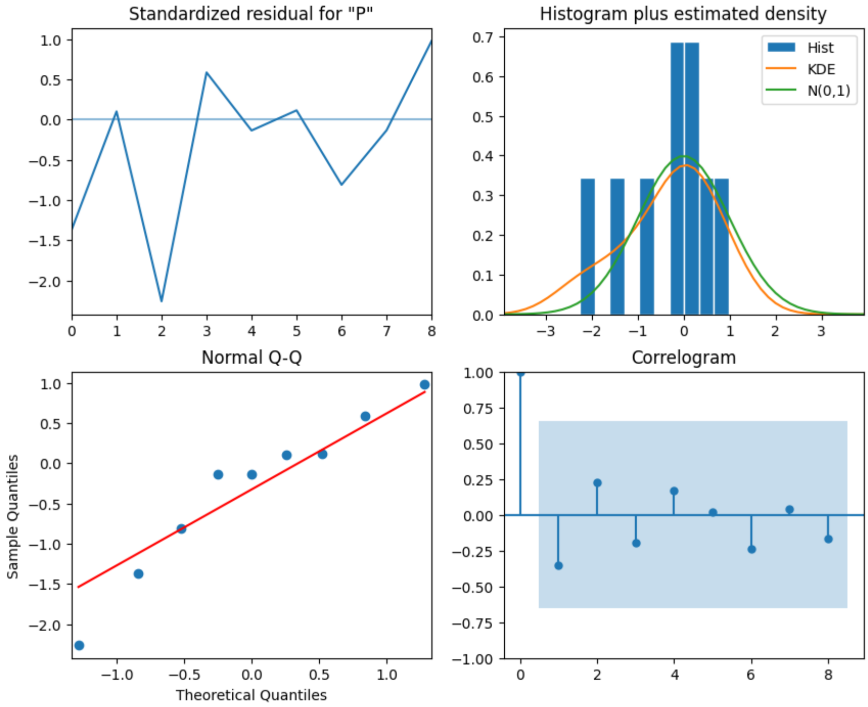

Figure 4: Diagnostic Plots for A Statistical Model

estimated density curves: -Histogram Bars: Show the frequency of residuals, Orange Line: Kernel Density Estimate (KDE) - a smoothed version of the histogram and Green Line: Represents the standard normal distribution (N(0,1)). The alignment of the orange and green lines with the histogram bars suggests whether the residuals follow a normal distribution.

Normal Q-Q Plot

A Quantile-Quantile (Q-Q) plot compares the sample quantiles of the residuals to the theoretical quantiles of a standard normal distribution:Red Line: Represents the expected line for normally distributed residuals and Points: Represent the actual residuals. The closer the points are to the red line, the more normally distributed the residuals are.

Correlogram

This plot shows the autocorrelation of the residuals at different lags: Points with Error Bars: Indicate the correlation values at various lags and the Shaded Area:Represents the confidence interval. Values within the shaded area suggest no significant autocorrelation, indicating the residuals are independent over time.

Table 3 shows the results of a Dickey-Fuller test, which is used to test for the presence of a unit root in a time series sample. The test statistic of -5.3314 is more negative than all the critical values at the 1%, 5%, and 10% significance levels. Combined with the very low p-value, this provides strong evidence to reject the null hypothesis. This suggests that the time series is stationary and does not have a unit root.

Chart 5 present the times series plots of the private sector over the last 12 years indicating that the AUM of private pensions has always been on the upward trajectory.

Table 4 presents on the forcasted times series values of the private sector AUM from 2012 to 2040 taking into account a 95% confidence intervals for both the lower and upper bound.

Table 3: Dickey-Fuller Test Result

Metric	**Value**
Test Statistic	-5.331440528065261
p-value	4.711563618849688e-06
# Lags Used	2.0
Number of Observations Used	9.0
Critical Value (1%)	-4.473135048010974
Critical Value (5%)	-3.28988060356653
Critical Value (10%)	-2.7723823456790124

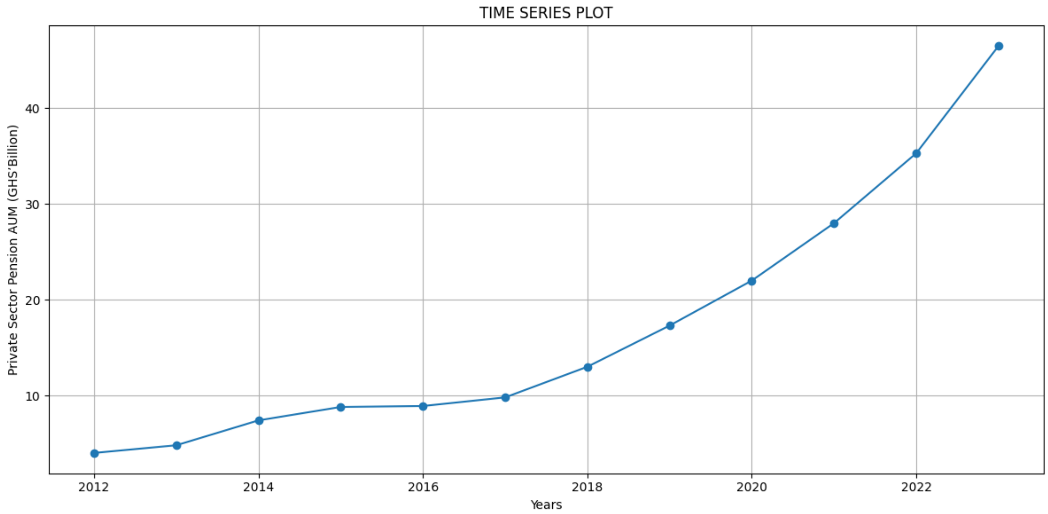

Figure 5: Time Series Plot

Table 4: Forcasted times series values of the private sector AUM from 2012 to 2040

Year	Forecasted Values (GHS'Billion)	Lower Bound (95% CI) (GHS'Billion)	Upper Bound (95% CI) (GHS'Billion)
2013	5.423536852	3.275484696	7.571589008
2014	6.720661869	-0.319161914	13.76048565
2015	7.247450909	-6.312625896	20.80752771
2016	8.354215719	-13.60046814	30.30889958
2017	11.15636763	-22.46023611	44.77297137
2018	15.24192579	-34.48246896	64.96632053
2019	19.61959785	-50.78297042	90.02216611
2020	24.0972536	-71.21520887	119.4097161
2021	29.28331686	-95.25346974	153.8201035
2022	35.6109406	-123.0317088	194.25359
2023	42.86059712	-155.3054794	241.0266737
2024	50.61826353	-192.6798141	293.9163411
2025	58.83731515	-235.2233702	352.8980005
2026	67.78828048	-282.8182577	418.3948187
2027	77.63626747	-335.6172463	490.8897813
2028	88.27024515	-394.0358105	570.5763008
2029	99.51904349	-458.4250597	657.4631466
2030	111.3777465	-528.9098723	751.6653653
2031	123.9652102	-605.5382018	853.4686221
2032	137.3429456	-688.4720041	963.1578953
2033	151.4567871	-777.979251	1080.892825
2034	166.2367155	-874.2954576	1206.768889
2035	181.6869161	-977.5598355	1340.933668
2036	197.8590336	-1087.881322	1483.599389
2037	214.7753175	-1205.417149	1634.967784
2038	232.4100458	-1330.365886	1795.185977
2039	250.7348986	-1462.908715	1964.378512
2040	269.7542712	-1603.185231	2142.693774

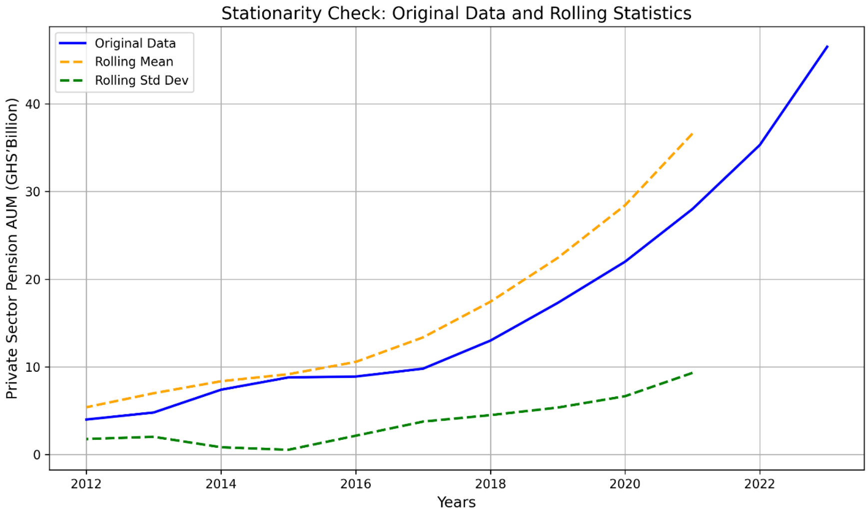

Figure 6: Stationarity Check: Original data and rolling statistics

CONCLUSION

The ARIMA (1,1,0) model provides a reasonably good fit for the Private Sector Pension AUM series. The statistically significant coefficient of the autoregressive term AR (1) indicates that current values are strongly influenced by their immediate past observations. Model diagnostics further show that the residuals are free from serious autocorrelation, as confirmed by the Ljung-Box test, and are approximately normally distributed according to the Jarque-Bera test. Nonetheless, signs of heteroskedasticity were detected, which suggests the need for additional adjustments to enhance the model's robustness.

In terms of accuracy, the model yields acceptable forecast error measures, including MAE, RMSE, and MAPE, making it suitable for short-term predictions. Moreover, the Augmented Dickey-Fuller test confirms the stationarity of the series, meaning its statistical properties such as mean and variance remain stable over time. This stationarity is particularly important, as it provides a strong foundation for reliable time series modeling and forecasting.

Finally, we suggest extensions to the model (ARIMA with GARCH, SARIMA with GARCH) to specifically tackle the heteroskedasticity problem for future research work

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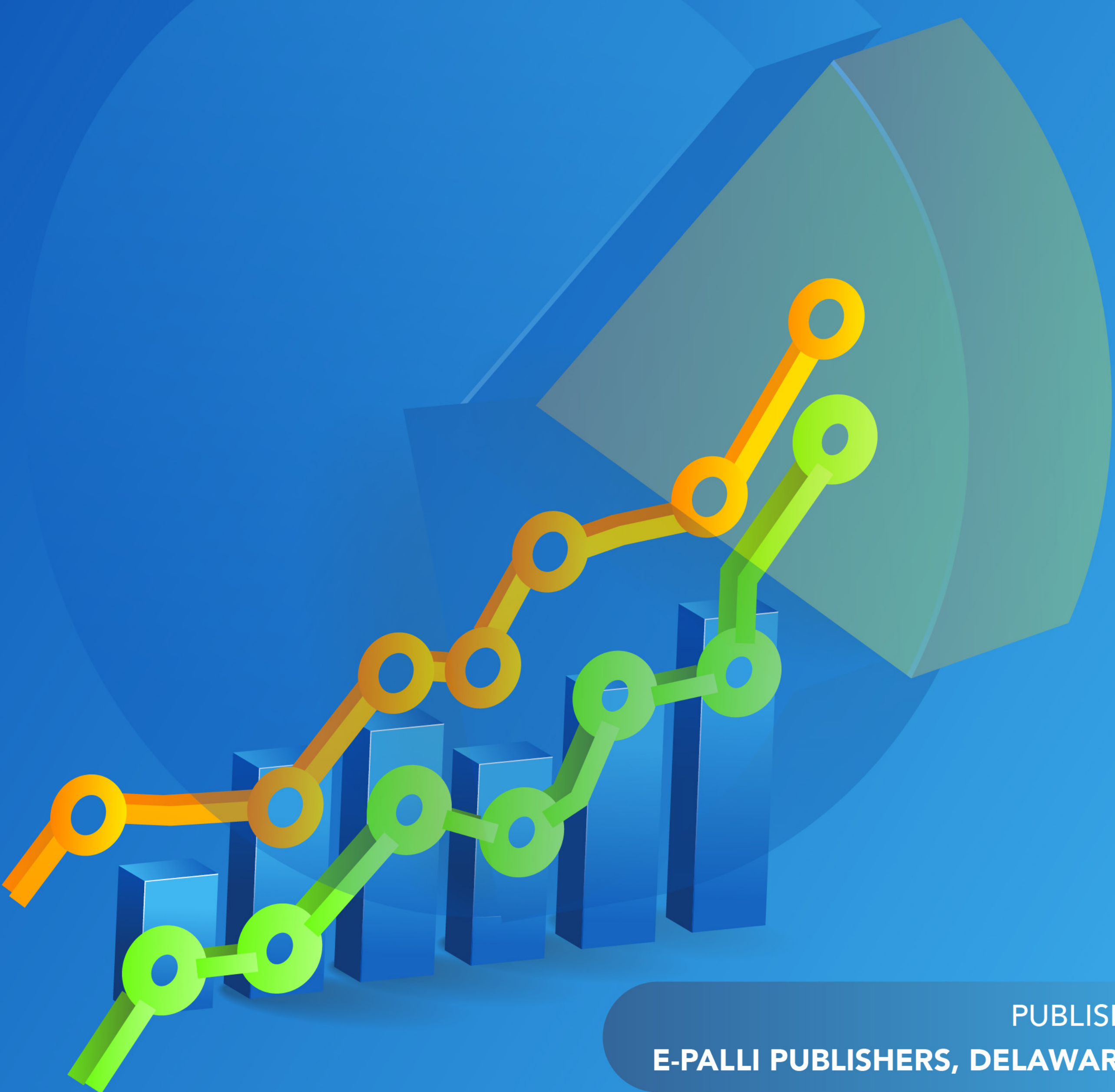

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Forecasting Key Macroeconomic Indicators in Ghana Using a Time-Varying VECM with Conformal Prediction Intervals

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Conformal Prediction, Exchange Rate Forecasting, Ghana, Inflation, Monetary Policy, Time-Varying VECM

ABSTRACT

This paper presents a six-month-ahead forecast of three key Ghanaian macroeconomic indicators: the USD/GHS exchange rate, the Consumer Price Index (CPI) and the Monetary Policy Rate (MPR). A Time-Varying Vector Error Correction Model (TV-VECM) is utilized to capture dynamic interrelationships among the variables. Conformal prediction intervals are incorporated to quantify uncertainty under minimal distributional assumptions. The results suggest moderate currency depreciation, persistent inflationary trends and stability in nominal interest rates over the forecast horizon.

INTRODUCTION

Forecasting macroeconomic indicators is essential for effective policy design and economic planning. This study employs a Time-Varying Vector Error Correction Model (TV-VECM) to analyze short-term movements in three critical Ghanaian macroeconomic variables: the USD/GHS exchange rate, the Consumer Price Index (CPI) and the Monetary Policy Rate (MPR). The TV-VECM framework captures both long-run equilibrium and evolving short-term dynamics. To account for forecast uncertainty, conformal prediction intervals are utilized which offer valid coverage without assuming specific error distributions.

LITERATURE REVIEW

Time-varying VECM (TV-VECM) for Ghana

Macroeconomic relationships in Ghana among inflation, the cedi/US$ rate, policy rates, money, output, and commodity prices are well known to be non-stationary with evolving long-run equilibria and shifting short-run dynamics (policy regime shifts, commodity price cycles, IMF programs, and disinflation episodes). Standard (time-invariant) VAR/VECM studies on Ghana capture cointegration and error-correction but assume fixed parameters, which can miss structural drifts that matter for forecasting and policy analysis. Using a time-varying cointegration framework lets the data accommodate gradual changes in adjustment speeds or even the cointegration vector itself exactly the type of flexibility needed in an economy that has seen alternating easing cycles and large disinflation in 2024–2025.

Core Advances on Time-Varying Cointegration and VECM

Early contributions showed how cointegration can evolve smoothly and proposed tests for time-invariance of the long-run vector (and rank). Bierens & Martins (2009/2010) formalize a time-varying cointegration setup where the cointegrating relationship changes smoothly; they derive a likelihood-ratio test against time-variation. This line of work motivates allowing βt and even αt to drift in a VECM.

A cautionary strand highlights pitfalls with state-space/Kalman estimation of time-varying cointegration: if not handled carefully, the Kalman filter can absorb unit-root behavior into the time-varying state and spuriously "find" time-varying cointegration between unrelated I(1) series. Robust procedures and bootstrap testing frameworks are proposed to distinguish no cointegration vs. fixed vs. time-varying cointegration. This is highly relevant if one estimates TV-VECMs for Ghana with state-space methods.

Recent econometric theory pushes further with smoothly time-varying VECMs: Gao, Peng & Yan (2023, 2025) develop a time-varying Granger Representation Theorem, estimation/inference for both short-run and long-run coefficients, a singular-value-ratio rank test, and stability tests providing a principled toolkit to build and validate TV-VECMs without resorting solely to ad-hoc filters.

Related strands include threshold/smooth-transition VECMs (to handle nonlinear adjustment) and applications showing that allowing for time-variation materially changes conclusions about long-run relations. These reinforce the empirical gains from flexible cointegration structures.

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Estimation Strategies for TV-VECMs

Three broad routes appear in the literature:

State-space / Kalman TVP-VECMs (or TVP-VARs with Cointegration)

Flexible but require care to avoid spurious cointegration; bootstrap or robust testing is recommended.

Nonparametric/Smoothly-Varying Coefficients

Treat $\alpha(\tau)$, $\Pi(\tau)$ as smooth functions of rescaled time τ with associated theory for rank, stability, and inference; this avoids some Kalman pitfalls and aligns with gradual Ghanaian regime shifts.

Bayesian TVP-VAR/Cointegration Frameworks (And Sparsity Priors)

Common in macro; while much of this work is in TVP-VARs, ideas translate to VECMs (shrinkage on time-variation, stochastic volatility), though dedicated Bayesian TV-VECM software remains less standard.

Forecast Uncertainty

Classical VECM forecast intervals rely on parametric assumptions (Gaussian errors, correct specification). In volatile, shifting environments like Ghana's, distribution-free uncertainty quantification is attractive. Conformal prediction (CP) provides finite-sample, model-agnostic prediction sets with marginal coverage guarantees. For time series, Adaptive Conformal Inference (ACI) and related online methods adjust interval widths to distribution shift and non-exchangeability, which are precisely the challenges in macro data with evolving regimes.

Key extensions handle multi-step and multivariate time-series forecasting crucial when producing joint paths for inflation, FX, and policy rates or when reporting horizon-h TV-VECM forecasts. Recent work develops online/multi-step CP with provable properties, and multivariate CP to form valid prediction regions across series.

An alternative, EnbPI (Ensemble Batch Prediction Intervals), uses ensemble residuals to deliver approximately valid intervals under dependence and shift; while often paired with ML forecasters, it is model-agnostic and can calibrate VECM residuals too.

TV-VECM and Conformal Pipeline for Ghana

The literature jointly suggests:

Specify a Ghana macro system [P_t, e_t, i_t, y_t, m_t]

Inflation, exchange rate, policy rate, output proxy, money; optionally commodity/terms-of-trade).

Estimate Cointegration And Time-Variation

Using a smoothly time-varying VECM (Gao–Peng–Yan framework) or carefully designed state-space TV-VECM with robust cointegration testing/bootstrapping to guard against spurious time-variation (Eroğlu *et al.*, 2022).

Forecast

Multi-step paths from the TV-VEC

Wrap Forecasts with CP

use online ACI (or variants) for one- and multi-step horizons; for multiple variables/horizons, apply recent multivariate/multistep conformal methods to obtain valid joint or per-horizon intervals that adapt to regime changes particularly important around policy turning points documented for Ghana in 2025.

Empirical Expectations And Gaps

Relative to fixed-parameter VECMs used in many country studies, a TV-VECM should improve calibration during regime shifts and commodity shocks, and better capture changing error-correction speeds.

Conformal layers are complementary to econometric inference: they provide finite-sample predictive coverage without re-specifying the TV-VECM and remain robust to mild misspecification.

Gap: Few (if any) studies combine TV-VECM with conformal intervals in macroeconomic practice especially for Ghana. The emerging multi-step/multivariate CP literature now makes this feasible and methodologically justified.

MATERIALS AND METHODS

Data Description

Monthly data from January 2014 to June 2025 were compiled. CPI data were obtained from the Ghana Statistical Service, while exchange rate and MPR data were sourced from the Bank of Ghana. All series were tested for unit roots using the Augmented Dickey-Fuller (ADF) test and found to be integrated of order one, I(1). Cointegration was verified using Johansen's method, confirming at least one cointegrating relationship among the variables.

The ADF test is based on the following regression:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta y_{t-i} \quad (1)$$

The Johansen cointegration test is derived from the Vector Autoregression (VAR) representation:

$$\Delta X_t = \Pi X_{t-1} + \sum_{i=1}^{k-1} \Gamma_i \Delta X_{t-i} + \epsilon_t \quad (2)$$

where the rank of the matrix Π determines the number of cointegrating relationships.

Model Specification

A Time-Varying VECM was implemented using a rolling window approach. This technique allows model parameters to evolve, accommodating structural breaks and time-varying relationships. The model captures both equilibrium correction mechanisms and short-term adjustments.

Forecasting and Conformal Prediction

A six-month forecast horizon was selected. To quantify forecast uncertainty, conformal prediction intervals were constructed using residual-based nonconformity scores. These intervals maintain valid coverage under mild assumptions and are robust to model misspecification.

Gradient Boosting Regression was employed to generate point forecasts. This method builds an ensemble of decision trees in a forward stage-wise fashion, minimizing a loss function by iteratively fitting residuals:

$$F_0(x) = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^{n} L(y_i, \gamma) \tag{3}$$

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x) \tag{4}$$

$$h_m(x) = -\left[\frac{\partial L(y, F(x))}{\partial F(x)}\right]_{F(x)=F_{m-1}(x)} \tag{5}$$

RESULTS AND DISCUSSION

Table 1: Forecast and 90% Conformal Prediction Intervals

Date	USD/GHS	[PI]	CPI	[PI]	MPR	[PI]
2025-07-31	10.4600	[10.12, 10.90]	258.87	[254.09, 261.07]	28.00	[28.00, 28.05]
2025-08-31	10.5900	[10.19, 10.93]	261.36	[260.11, 262.67]	28.00	[28.00, 28.01]
2025-09-30	10.7100	[10.42, 11.07]	264.56	[262.61, 265.19]	28.00	[28.00, 28.01]
2025-10-31	10.8400	[10.55, 11.21]	268.25	[265.82, 268.43]	28.00	[28.00, 28.01]
2025-11-30	10.9700	[10.67, 11.35]	272.09	[269.53, 272.18]	28.00	[28.00, 28.01]
2025-12-31	11.1000	[10.80, 11.48]	275.99	[273.39, 276.08]	28.00	[28.00, 28.01]

Source: Authors Computation and Projections

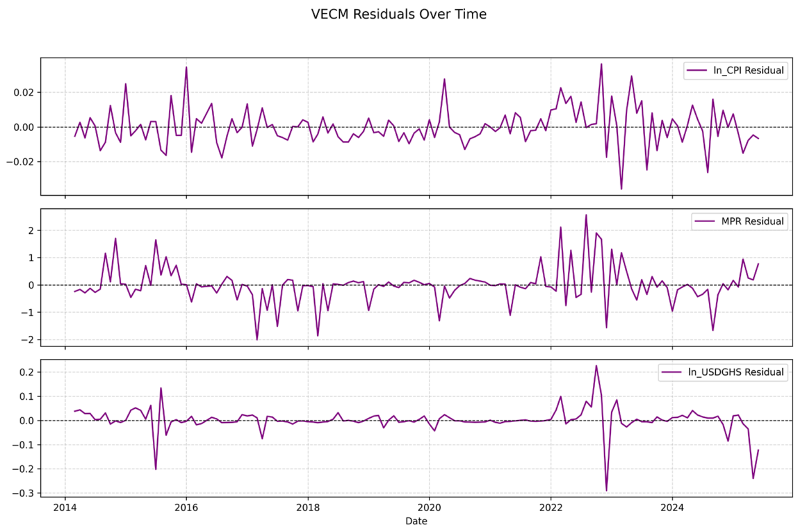

Figure 1: Residual plot for ln CPI, MPR, and ln USDGHS (July–December 2025)

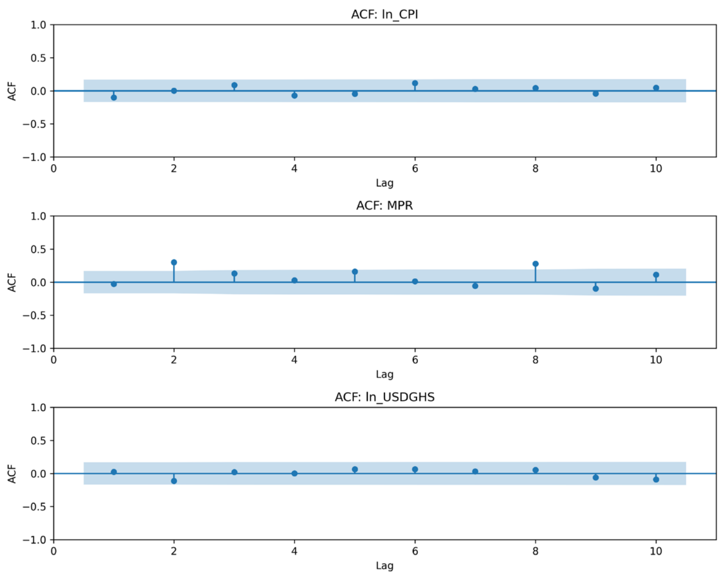

Figure 2: Autocorrelation Function (ACF) plot for residuals

Residual Plot Observations

• ln_CPI: Residuals are centered around zero with low variance but show a slight increase in variance from 2022 onward.

• MPR: High residual variance with frequent spikes, suggesting potential outliers or structural breaks.

• ln_USDGHS: Residuals are mostly stable over time, with a few sharp spikes between 2022 and 2023 likely reflecting volatility during that period.

Autocorrelation Function (ACF) Observations

• ln_CPI Residuals: All spikes are within the 95% confidence bands across lags 0 to 10, confirming the absence of significant autocorrelation.

• MPR Residuals: A visible spike at lag 2 exceeds the confidence bounds, indicating residual autocorrelation.

• ln_USDGHS Residuals: All spikes remain within the confidence bounds, confirming white noise residuals.

Discussion

USD/GHS Exchange Rate

The forecasted exchange rate path indicates a gradual depreciation of the Ghanaian Cedi, rising from 10.46 to 11.10 over the six-month period. The widening prediction intervals over time reflect increasing uncertainty as the forecast horizon extends.

Consumer Price Index

CPI is projected to increase gradually from 258.87 to 275.99, suggesting continued inflationary pressure. The forecast intervals remain tight, indicating strong model confidence in the inflation trajectory over the horizon.

Monetary Policy Rate

The MPR remains effectively stable at around 28.00% across the entire forecast window. The extremely narrow interval bounds suggest little model uncertainty and high temporal persistence in this rate.

Model Pitfalls and Deployment

From the estimated parameters value and diagnostic plots, we observed that Monetary Policy Rate residuals show strong autocorrelation and large spikes especially after 2022, confirming that the model does not capture MPR dynamics well. This suggests the presence of possible outliers, structural breaks after 2022, and insufficient lags. Therefore, the model can be deployed to forecast only the Consumer Price Index (CPI) and USD/GHS. The long-run links look usable, but MPR short-term forecasts may not be reliable unless the MPR pitfalls are fixed and improved.

CONCLUSION

This study utilizes a Time-Varying VECM with conformal prediction to forecast Ghana's key macroeconomic indicators over a six-month horizon. Results suggest moderate exchange rate depreciation, persistent inflation, and stability in nominal interest rates. These forecasts are grounded in historical data patterns and subject to limitations arising from policy shocks or structural changes outside the model's framework.

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Investigating the Factors that Affect the Financial Performance of Ghana's Insurance Companies

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Abstract:- The financial business is rapidly expanding, allowing for high quantities of transactions. This expansion has raised the demand for insurance and insurance products dramatically. Though previous research has looked at the elements that drive the profitability of the insurance business from both a general and a life viewpoint, little attention has been paid to the main indicators that define the financial performance of these insurance organizations. This study looked at the factors that impact the financial success of insurance businesses in Ghana, a developing country. It investigated the elements that influence financial performance. The findings of the study showed that General insurance was the most purchased type of insurance policy. It was also discovered that the majority of the policyholders who buy insurance policies were 24 years and above.Finally, the study showed that the most influential factor considered for the financial performance of the insurance company is the effective management factor followed by the competition factor. Other factors that were also identified were personality influence, liquidity, and profitability. As a result, the purpose of this project is to raise knowledge about the variables that impact the financial performance of insurance businesses in Ghana. This knowledge, on the other hand, will assist policyholders, stakeholders, brokers, and others concerned in understanding their insurance company's financial status and the variables impacting it. This research would also provide knowledge on the influential factors for insurance companies' financial performance as well as the most purchased insurance policy, which would help insurance companies understand why policyholders and investors patronize insurance by making more improvements to ensure a higher insurance penetration in Ghana.

Keywords: *General Insurance, Insurance Policy and Penetration, Factor Analysis, Effective Management, Liquidity, Profitability and Personality Influence.*

I. INTRODUCTION

The insurance industry contributes positively to Ghana's economy. Insurance businesses, as part of a country's overall financial system, provide great financial services to support economic growth and development. The insurance of risks inherent in economic organizations and the mobilization of a large number of money through premiums for long-term investments are examples of specialized financial services (Agiobenebo and Ezirim, 2002). There is extremely little awareness about the insurance sector, particularly in a developing market like Ghana.Insurance works on the premise of risk pooling (loss pooling), in which people pay to a common fund in the form of premiums, and the fortunate who do not experience loss assist the unlucky who do suffer loss within a set insurance term. Insurance firms provide policyholders coverage for many sorts of risks, such as commercial property damage or loss, health and casualty, monetary losses, fire and theft, motor vehicle accident, and so on. The risk-absorption role of insurers fosters financial stability in financial markets and gives economic entities a sense of security.

Without insurance, the corporate world is unsustainable since hazardous enterprises may not be able to retain all types of risks in this ever-changing and uncertain global market (Ahmed, Ahmed, and Ahmed, 2010). Insurance firms get a premium from policyholders in exchange for this risk protection, which is used to pay expenditures and the predicted risk if they occur. The ability of an insurance business to continue covering risk in the economy is dependent on its ability to generate profit or value for its stakeholders.

A well-developed and evolved insurance business is advantageous for economic growth since it offers long-term finances for each economy's infrastructure development (Charumathi, 2012). Insurance firms can also provide long-term savings that can be utilized to support projects with extended maturities. The viability of the insurance sector is critical to any economy. In an economy, there is a clear relationship between sustainability and the financial performance of insurance businesses. Institutional stakeholders such as insurers, pension trusts, and sovereign wealth funds have more than USD 80 trillion in assets under management globally that can be deployed to fund long-term projects (PwC AWM Research Centre, 2017. Pp.6-7).

Consumers, investors, and policyholders should all be concerned about the insurer's financial sustainability and ability to make its claim payment commitments to policyholders. Part of the premiums is invested to get better yields for longer-term risk protection. Though the protection buyer transfers individual risk to the insurer's larger and

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more diverse portfolio, the risk is not eliminated because the insurer may default on his commitments. When the losses on the diverse portfolio exceed the predicted losses, insurers must have enough equity or buffer capital to pay their commitments. Due to a lack of experienced insurance salesmen to offer goods, the general public is left with no foundation to make an educated expenditure or investment choice, or to choose which business to place their cover with (Kumba, 2011). There are many factors to examine when looking at the performance of insurance companies such as marketability, competition, profitability, and others. This research pursues to set the pace for more scientific research and also for academic purposes into the general financial performance of insurers in Ghana.

According to Shawar (2019), the financial performance of every sector in any country acts as a cornerstone of economic growth and industrialization. Insurance firms' financial performance may be assessed by identifying both internal and external elements represented by specific fundamental attributes of an insurance company. Under the financial industry's umbrella, the insurance sector plays a critical role in regulating money to many businesses, consequently delivering significant inflows to economic and financial growth. Furthermore, the insurance industry is deemed well-established if it can ameliorate any type of financial crises in the economy, hence boosting the country's economic structure. Financial performance is basically how a company uses its resources to generate a profit, and it is assessed using return on assets, return on sales, and growth. According to Jim (2007), the performance of any monetary institution can be assessed in a variety of ways, including profit growth, employee growth, asset growth, or any other sort of variable factor that management believes is a significant producer of a business's anticipated success.

The majority of life insurance performance research has focused on discovering insurer-specific characteristics and indicators that help in identifying insurers that are more likely to go bankrupt. Deakin (2005) investigated the 'run on the bank risk' and discovered that before 1992, rating firms did not usually recognize the risks contained in liabilities such as guaranteed investment contracts.

Furthermore, due to an apparent lack of uniform financial reporting formats, several insurance companies have failed to publish their profit and loss accounts, and some insurance companies have also failed to submit their audited annual report within the statutory time frame, making it difficult for the general public to gauge their profitability, overall written premiums, or even net incomes. Some corporations' sparse financial statements, given to the public, leave a lot of grey areas and possibilities for unethical behavior such as tax evasion and concealment of important ratios and numbers.

The insurance industry in Ghana; both Life and Non-Life insurance companies. Amid Ghana's present crisis, the insurance business has seen massive regulatory reforms and technological developments. Insurance is intended to transfer unanticipated losses by transferring such risk to the insurance provider, which compensates the policyholder in the event of unanticipated losses. The notion of insurance first appeared in Ghana in 1955, and the industry is growing more active and competitive, with significant development made between 2015 and 2019. The office of the Commissioner of insurance was established under these provisions to strengthen the government regulation on insurance. The Commissioner of insurance was created as a department under the ministry of finance. To enhance the supervisory capacity of the regulator, the government delinked the department from the ministry to give it some autonomy.

The Insurance Act number 724 of 2006 established the National Insurance Commission with the commissioner of insurance as the managing director and the chief executive officer to take the role of regulating, supervising, and developing the insurance industry. The role of the Commission is: to ensure effective administration, supervision, regulation, and control of the insurance and reinsurance business in Ghana, to protect the 6 interests of insurance policyholders and insurance beneficiaries in any contract of insurance, to promote the development of the insurance sector and to advise the government on the national policy to be followed to ensure adequate insurance protection and security for national assets and national properties among other functions.

- *New Insurance Act:*

Given the changes in the insurance landscape, the NIC was evaluating the Insurance Act to make it more appropriate. The new Insurance law was enacted in 2021, and it was consistent with the most recent Insurance Core Principles of the International Association of Insurance Supervisors. It is also a primary duty of the Commission to verify practitioners' conformance to globally accepted standards. The NIC's latest frameworks, directives, and guidelines were included into the new Insurance Act.. The enactment of Act 1061 was a major milestone towards a robust insurance regulatory environment as it empowers and grants adequate powers to the Commission. The insurance market of Ghana is relatively stable with more prospects and opportunities to explore. The total profit that was declared by the insurance industry by the end of 2019 was GHS 196 million and that of total corporate tax was GHS 72 million. The total number of people employed in the industry as an asset in 2019 was 12,000. According to the insurance industry report for the year 2019, there were 150 licensed insurance companies at the end of 2019. The total asset of the insurance industry according to NIC 2019 was GHS7.65 billion. It is against this background that, the NIC seeks to explore the challenges and strategies for increasing insurance penetration in line with the objective of the Commission.

- *Research Problem:*

Andersson (2010) noted that a country's economic growth is heavily reliant on a healthy financial system, and that without a strong insurance industry, the economy and the wealth creation connected with it can suffer (International Accounting Standards Board, 2007). The insurance business is an essential component of the country's financial sector, and its advantages cannot be overstated. Ghana's non-life insurance industry is one of the fastest growing in Sub-Saharan Africa, according to Fitch Solutions (2021), and it is evolving.If this crucial sector was missing, the consequence on the economy would be devastating, knocking off billions of cedis from the Gross Domestic Product (GDP) index.

However, although offering crucial intervention policies and producing wealth via investments, the insurance business in Ghana and other nations has had its fair share of firm failures (Greene, 2000 Hagel, Brown, & Davison, 2010). The Bank of Ghana (BoG) and the Securities and Exchange Commission (SEC) revoked the licenses of hundreds of financial organizations in 2019, including savings and loan businesses, microfinance companies, finance houses, investment management firms, and other non-bank financial institutions. This had a huge impact on the solvency and liquidity of insurance businesses since a large number of regulated insurance companies had significant investments with these institutions. Several insurance firms have gone bankrupt or been dissolved in the previous decade. The majority of these enterprises have failed, leaving policyholders, pension plans, and life funds with billions of Cedis in cash. This brings out the question of whether insurance companies are financially sound and whether they are disclosing enough information to enable investors to make informed decisions. Financial health is critical for any business organization.

Previous empirical research on the factors influencing the financial success of the insurance sector has mostly focused on the United States, Europe, Asia, and East Africa. To establish the financial performance of the insurance business, several of these research focused solely on company-specific characteristics such as firm size, age, leverage, net premium earned, and profitability. Hardwick and Adams (2002) conducted a research on the UK life insurance business, as did Makembo (1992), who explored flaws in Kenya's compensation system (fault system) for personal injuries and fatalities in automobile insurance.

According to Greene (1963), most important research on the difficulties confronting the sector in Africa have focused on people's attitudes and views about insurance. Among these studies, Yusuf et al. (2009) in Nigeria and Ackah and Owusu (2012) in Ghana are noteworthy.

However, no empirical or particular study has been conducted on the elements that impact the financial performance of insurance businesses in Ghana. The study aims to bridge a knowledge vacuum in this area. The study intended to determine whether and how certain elements impact the performance of insurance businesses in Ghana, as well as which of the factors is most significant as a financial performance indicator.

➢ *Objectives of the Study:*

The main objective of this study is to investigate the factors that influence the financial performance of insurance companies in Ghana. To achieve the main objective of the study, the following specific objectives are set:

- To identify significant factors influencing the financial performance of insurance companies in Ghana
- To identify the most influential factor that influences the financial performance of insurance companies in Ghana.
- To identify the most purchased insurance type policy.

➢ *Research Questions:*

Given the aforementioned issue and the perceived advantages of insurance firms, the following research questions need careful examination as a means of assessing the elements that impact the financial performance of insurance companies in Ghana.

- What are some of the key factors that determine the financial performance of insurance companies in Ghana?
- What are the types of insurance policies that policyholders mostly buy?
- What is the most influential factor influencing financial performance in insurance companies?

II. LITERATURE REVIEW AND THEORITICAL FRAMEWORK

According to Chang Lee and Chang (2013), the average rate of growth in the insurance business (i.e., 10%) was substantially greater than the worldwide rate of economic growth. The insurance sector is an essential component of the global financial market, with insurance companies serving as major institutional investors. Many current researchers have emphasized the significance of financial institutions in a country's development (Catalan, Impavido&Musalem, 2000); Levine, (1992); Nguyen, (2020).According to Saunders and Cornet (2008), financial institutions can be classified as commercial banks, savings banks, or insurance businesses. The bulk of them, however, highlight the function of the stock market or banks as the primary financial institution that may encourage economic progress. According to Ward and Zurbruegg (2000), insurance not only transfers and indemnifies risk, but it also serves the function of other financial institutions, such as if insurance has been poorly investigated (Hasna 2010; Haiss&Sumegi, 2008). Because of the critical necessity of insurance in a country's entire financial system, the function of insurance has grown throughout time.Nonetheless, the research on the factors influencing insurance financial performance is compared to other financial organizations. Expanding on the relationship between GDP and insurance market growth, it is important to recall that the basic job of the insurance sector is to provide people and companies with coverage against certain events by spreading losses among the pool of policyholders. As a result, insurance firms underwrite, manage, and finance risks. Furthermore, insurance is employed by businesses to hedge corporate risk, and it cannot be utilized for speculative objectives like derivatives.

According to NIC (2019), the principle of finance theories include understanding the procedures by which organizations and people generate cash or capital for any type of spending, as well as how money is distributed to projects while taking into account the risk aspects involved. It also involves the study of money and other assets, the management and profile of project risks, asset control and management, and the science of money management. This theory is used to model human decision-making, particularly in the context of microeconomics, where it assists economists in better understanding a society's behavior in terms of individual actions as explained by rationality, in which choices are consistent because they are made based on personal preference. The term rationality is used differently depending on the issue. Many additional ideas and concepts are involved with the market mechanism that allows for the creation and distribution of products. However, the Rational Choice Theory is a foundational game theory that gives a comprehensive mathematical framework for studying individuals' mutually interdependent interactions including resources such as prestige, time, and many others. According to the Rational Choice Theory, humans are driven by their own goals and preferences. Human behaviors are heavily influenced by knowledge about the conditions in which a specific individual will operate and attempt to achieve his or her objective. The selection of goals, as well as the selection of an appropriate technique to accomplish the previously stated aim, is critical in the domain of Rational Choice Theory. It regards reward and punishment as benefits and costs, respectively, and the theory contends that human behavior is governed by the desire to obtain favorable rewards (Organization for Economic Cooperation and Development, 2003). This hypothesis is applicable in the insurance sector. The reasonable decision for businesses is based on numerous criteria, such as the insurance company's financial success metrics. Most people would not acquire policies if they knew their prospective insurance company was likely to fail since, in the event of a risk, they would not receive benefits from the policies they purchased.

- *Interest Rate:*

Higher real interest rates have an uncertain influence on both life and non-life premiums. Because insurance firms make pledges or obligations to policyholders at the time of policy sale, they are not free to modify the rates specified or agreed in the sale later based on situation. This feature of insurance exposes them directly to the dangers associated with interest rate increases. Because insurance firms invest a large portion of their collected premiums, the money gained by investing is strongly dependent on interest rates. According to Beck and Webb (2003), higher real interest rates would boost the investment return of providers, allowing them to provide more appealing returns to customers. In a comprehensive research of the Chilean annuities market, Rocha and Thorburn (2007) and Rocha et al (2008) show that an increase in real interest rates has a favorable influence on real annuity rates but an equivocal effect on the quantity of new annuity contracts and the annuity premium.

Another disadvantage of interest rate swings (which is not limited to insurance businesses) is the cost of borrowing. A high-interest rate has a significant negative impact on the premium charged to policyholders, making it difficult for many people to purchase insurance products; however, if the rate charged is consistent, low, and nondiscriminatory, there will be a high penetration of insurance accessibility. For example, if all policyholders have the same risk exposure units, the same premium should be charged regardless of whether the cost of borrowing is high. Best (1992) discovered that the number of insolvencies is connected with the accident and health underwriting cycle in a study of the relationships between insurance market circumstances and insolvencies. The growing incidence of bankruptcies is partly tied to rising interest rates and the life-health insurance industry's emphasis on investment-related products. The Best research did not evaluate the numerous economic elements in a multivariate framework, making it impossible to determine the relative importance of the separate components.

Santomero and Babbel (1997) reported that in the United States, many life insurers' management failed to effectively foresee economic shocks (e.g., unfavorable interest rate movements), which harmed both business profitability and the rate of product-market growth. This insight implies that, if future economic shocks are mostly unexpected, so would business performance in the life insurance market. The asset/liability mismatch increases the insurer's leverage and increases the risk of bad performance (Carson and Hoyt, 1995), as well as guaranteed investment contract withdrawals (Carson & Scott, 1996).

- *Profitability:*

Mwangi (2013) proposed that profitability is a major driver for selecting whether to invest in his research of the elements that impact the financial performance of insurance firms. Ideally, you want to outperform the industry norm in terms of growth, but you also want to ensure that this higher growth does not come at the price of admitting higher-risk clients. A firm whose premium income is expanding at a slower rate, on the other hand, may be excessively choosy, seeking only the greatest quality insurance prospects. According to Santomero and Babbel (1997), the second area of profitability that should be considered is investment income.

Another important factor in assessing insurance business financial performance is the capacity of the insurance firm to pay claims to policyholders and give claim no discount to policyholders who do not make claims within a fiscal year. As a result, the effect of claim and award payment will have a long-term direct positive link with the financial success of the insurance business.

- *Competition:*

Efficiency is a popular quantitative indirect indicator of competition. High efficiency may imply the presence of competition and vice versa (Bikker&Leuvensteijn, 2008). The predominance of mergers and acquisitions among insurance carriers and agencies is one of the most significant developments in the insurance sector (Schich& Kikuchi, 2004). Economists have explored the measuring of industry rivalry in several ways over time. The first research sought to infer business competitive behavior and performance from industry market structure. The number of enterprises and any concentration of market share are thought to influence competitive behavior. Fewer enterprises with more concentrated market shares are more prone to engage in anticompetitive activities than a large number of small firms.

Blundell-Wignall, Atkinson, and Lee (2008) provided an alternate approach to competitive behavior and investigated company revenue and cost structures using the framework of perfect competition as a reference point. Firms in an industry with the perfect competition are unable to absorb any cost increases. They are required to pass on the whole increase in input costs in output pricing and income, but output remains unchanged. In contrast, under monopolistic circumstances in equilibrium, an increase in input prices, such as wages or administrative expenses, leads to a decrease in production and a price increase that is less than the increase in costs, resulting in a decrease in total income.

Firms with marginal profits may be forced to exit the sector. In long-run equilibrium, monopolistic competition determines output where the average cost curve is tangential to the average income curve. Because businesses produce at less than minimal cost, monopolistic competition theory argues that the industry has overcapacity. Firms in this field create money by insuring risks and investing their assets, according to Donlon and Gutfreund (1998). Given their underwriting losses, this indicates that enterprises in the GI industry must take more risks than would appear to be compatible with prudence. When applied to the topic at hand, this means that corporations can only recoup increased investment expenses by reshuffling their portfolios toward more er assets and thereby reaping larger returns. Market and credit concerns are manifestations of asset risk (O'Connor, 2000). Policyholders are also interested in the insurer's capacity to generate a positive product brand image and promote innovation in insurance service delivery. All of these variables have a beneficial impact on the financial success of the insurance sector. This is due to theindications in place, the insurance business will be able to benefit from the industry's major market share, resulting in a rise in the number of policyholders purchasing insurance from that firm.

➢ *Liquidity:*

Liquidity ratios are defined by Black, Wright, and Bachman (1998) as the amount of money that firms and other private organizations have on hand at any one moment to serve their obligation. Liquidity ratios are very significant when looking at a company's financial accounts and seeking to assess where it stands in terms of viability. The greater a company's liquidity ratio, the better. Entities having a high debt-to-liquidity ratio are more likely to fail and make riskier investments.

Liquidity risk could include two different types of risk: the risk that an insurance company will become unable to assure itself of adequate funding due to a decline in new premium income caused by a deterioration of its financial position, an increase in surrender value caused by large-lot cancellations, or an outflow of funds caused by a big disaster, or it will incur losses because it is forced to sell assets at markedly lower prices than normal and therefore unable to maintain cash flow and the risk that upheavals in the market will render it impossible to trade and therefore force the company to engage in transactions at prices that are markedly more disadvantageous than normal (Black, Wright & Bachman, 1998).

The acid test, according to Barney (1997), is the first measure of an insurer's ability to pay financial obligations. It determines if a company's short-term assets are sufficient to satisfy its immediate liabilities. When an insurer must sell assets early to meet claims, it suffers investment losses and hence poor financial performance. An insurer's cash flow should virtually always be positive. Cash flow is critical to an organization's survival. Having a sufficient amount of cash on hand ensures that creditors, workers, and others are paid on schedule. If a company or individual does not have enough cash to finance its activities, it is considered to be insolvent and, if the insolvency continues, it is a potential candidate for bankruptcy. Other factors to consider include the investment grades of the company's bond holdings. Too many high and medium-risk bonds may cause volatility and hence poor financial health.

➢ *Proximity:*

Proximity is a burgeoning theoretical framework in a variety of domains, including innovation studies, organizational science, and regional science (Knoben and Oerlemans, 2006). In organizational studies, proximity is a multidimensional notion that extends beyond geographical distance (Boschma, 2005). On the contrary, it refers to the degree of resemblance between individuals or organizations in aspects other than geographical proximity (Knoben and Oerlemans, 2006; Balland, 2012). A detailed examination of the notion of proximity is not only important for gaining a good understanding of the concept but it is also required due to its ambiguity. Economic, geographical, and sociological definitions of proximity vary and frequently pertain to conceptions of space, social ties, family, and institutions, to mention a few (Torre and Ballet, 2005; Dankbaar, 2007). As anticipated, proximity has gained a more abstract meaning, thus expanding from a being mere measure of geographical distance to encompassing several other dimensions (Knoben and Oerlemans, 2006).

As a result of these factors, as well as a thorough assessment of the relevant literature, proximity is defined as the degree of resemblance between entities in geographical, organizational, cognitive, social, institutional, and technical aspects. Proximity is defined as an organization's proximity to its customers and policyholders' places of residence, given that where insurance companies are concentrated and policyholders are widely dispersed, there is a tendency for a large number of policyholders and customers to buy insurance policies because of its numerous branches established at those places of residence. To that aim, it can be argued that the closer the insurance sector is to its policyholders, the better the financial performance of the insurance firms will be due to the huge number of policyholders purchasing an insurance policy.

- *Marketability:*

Marketing is defined by Kotler, Armstrong, Saunders, and Wong (2001) as a social and management process through which individuals and groups achieve what they need and desire by developing and trading products and values with one another. It is a critical managerial discipline that assures product and service makers can perceive consumer expectations and meet or exceed them. Because it covers the most significant components of marketability, the marketing process is critical to the commercial performance of insurance firms of all sizes.It is about comprehending the competitive marketplace and ensuring that you can capitalize on significant trends to reach consumers with the correct insurance product at the right price, location, and time. Clever marketing has resulted in several recent corporate success stories, ranging from large insurance and business-to-business firms to tiny, specialized players. As digital and mobile technologies permeate all parts of life, getting close has become more crucial than ever. This change has also given rise to new techniques for marketing that are more focused, relevant, and effective.

As competitive demands mount, marketing skills have never been more valuable. What was formerly considered a departmental activity is now considered a frontline business mindset for all employees. Professionals who develop and implement marketing strategies make a direct contribution to the economy. Their abilities attract and keep clients, raise sales, and enhance profits, resulting in a large number of customers for the insurance firm. The influx of marketability in the form of a television advertisement, awareness creation, and good customer relations have a positive impact on insurance financial performance because it brings in a large number of policyholders to buy insurance products, increasing the insurance industry's capacity to have access to a large pool of funds that can help them make claim payment when the proximate cause of the risk insured occurs.

- *Personality Influence:*

Personality and personality psychology has been researched since the early 1900s, according to Greer (2018), with Freud initiating the subject. This psychology grew in the Western atmosphere of individualism on the premise that each person is unique and hence has numerous distinctions from others, with personality being one of the defining features. Personality influence is a major component in determining whether or not to invest in any firm. Policyholders typically purchase insurance goods because specific types of people invest with such insurance organizations. Insurance exists because policyholders are compensated for their losses in the event of a risk, but with those personalities such as popular artists, footballers, Journalists, and families investing with those firms, they are assured of receiving their indemnity in the event of losses, which induces them to buy a large number of policies with the company. As a result, personality influences the financial success of insurance firms.

- *Empirical Review:*

Most earlier research on life insurance has focused on North America, Europe, and Asia, where markets may be classified as near-efficient, in contrast to markets in the developing world. According to Sigma (2001), the United States and Japan have the largest insurance sectors, accounting for more than half of worldwide premium income, followed by the United Kingdom, Germany, France, and Italy.Furthermore, throughout the previous four decades, the global insurance sector has exceeded global economic development on average. Between 1984 and 2001, the global insurance business expanded at a 9.7% annual pace (approximately 11.8 percent from the life insurance sector and 7.5 percent from the property-casualty sector). Growth in the global property-casualty market has slowed dramatically in recent years and has just kept pace with overall economic growth (Sigma, 2001).This development is primarily the result of a sustained downward trend in commercial business until profitability, interest rate fluctuations, liquidity, the financial performance of insurance companies, competition in the insurance industry, dependent variables, independent variables, and, most recently, the Asian economic and financial crisis, particularly in Japan, and the deregulation of the European market, both of which have resulted in appreciable price decreases.

This is in contrast to the life insurance industry, which has grown at a global rate of roughly 5.4% since 2000. The rising demand for private pension plans in the United States and Western Europe, as well as the soaring demand for unit-linked products, are primarily responsible for the growth in the life insurance market. In terms of total premiums, the Organization for Economic Cooperation and Development (OECD) nations accounted for 95.52 percent of the life insurance market in 1994 and 93.99 percent of the property-casualty premium volume in 2001, respectively. Furthermore, insurers in OECD nations have suffered declining premium revenue, decreasing capital market earnings, and low-interest rates, all of which have placed them under strain (Sigma, 2002). Furthermore, the expanding importance of the insurance business in emerging nations is evident in the increased insurance density and penetration of non-OECD insurance markets (Sigma, 1996, 2001).

Nonetheless, emerging economies have a long way to go before they can equal the relative size and importance of the insurance business in developed nations. Mwangi (2013) opined in a study on the factors that determine the financial performance of insurance companies in Kenya that the credit policy should specify credit risk philosophy governing the extent to which an institution is willing to accept credit risk; levels of authority should be subject to timely review to ensure that it remains appropriate to current market conditions and credit officers' expertise. The policy should be conveyed to all people associated with customer terms and payments, including sales representatives, for them to set the proper expectations with clients. Credit rules should also include criteria that allow the company to select the terms of sale that will be offered to the client.

III. METHODOLOGY

This paper employed the use of statistical techniques in the analysis of the data. Factor analysis is the main method used for the analysis of the data. The computer software used to analyze the data was the Statistical Package for the Social Sciences (SPSS) and Microsoft Excel Also, tables were used to support the analysis.

- *Data Collection:*

To meet the data requirement of this research, a questionnaire was designed as a quantitative study based on the researcher's pre-observations to find the most purchased insurance policy and the factors that influence the financial performance of insurance companies in Ghana. The study was conducted in three regions, namely, Greater Accra, Ashanti region, and Central region, which comprises top management and policyholders because the researchers live in these three main regions which happen to be the major areas where industrial insurance activities take place.

- *Sampling Techniques:*

Sampling is the process of choosing a subset of a population to represent the complete population. To collect data for the investigation, a straightforward sampling strategy was used. MacCallum (1999),states that the sample size required to get a greater probability is between 200 and 400 samples. This aids in successfully interpreting a large number of variables. In this case, 210 questionnaires were distributed to policyholders of a few insurance companies and 10 questionnaires were distributed to the top management of 5 insurance companies across the three regions.

- *Factor Analysis Model:*

let P variables be $X_1, \dots, X_p$ with the mean vector μ_p (which is assumed to be equal to zero and variance ε, then each of the observed variables can be written as a linear combination of m (m<p) factors as:

$$X_j = \gamma_{p1} f_1 + \gamma_{p2} f_2 + \dots + \gamma_{pm} + \varepsilon_p \text{ where j=1, 2…., p} \text{ ----------------- (1)}$$

Where γ_{ij} is the factor loadings of the i^{th} variable on the j^{th} factor;

f_i is the score of the common factor of the i^{th} observation;

ε_p are the specific factors for variable p

In matrix notation the factor model is represented as;

$$X = \lambda f + e, \quad \text{--(2)}$$

Where $f' = (f_1, f_2, f_3, \dots, f_m)$

$e' = (e_1, e_2, e_3, \dots, e_p)$

and,

$$\lambda = \begin{pmatrix} \lambda_{11} & \lambda_{12} & \dots \lambda_{1m} \\ \lambda_{21} & \lambda_{22} & \dots \lambda_{2m} \\ \vdots & \vdots & \vdots \\ \lambda p_1 & \lambda p_2 & \dots \lambda p_m \end{pmatrix}$$

- *The Communality of a Variable:*

The communality of a variable is the proportion of each variable's variance that can be explained by the factors. It is also denoted as h^2 and can be defined as the sum of squared factor loadings for the variables. The communalities for the i^{th} variables are computed by taking the sum of the squared loadings for that variable (McGarigal et al, 2013). This is expressed as:

$$\hat{h}_i^{\ 2} = \sum_{j=1}^{m} \tilde{\iota}_{ij}^2 \quad \text{------------------------- (3)}$$

Where $\hat{h}_i$ is the variability of the i^{th} variable

$\tilde{\iota}_{ij}$ is the loading or correlation between the i^{th} component and the j^{th} variable

- *Suitability of Factor Analysis:*

Two statistical tests must be conducted to examine the adequacy of the sample and the suitability of data for Factor Analysis. These are;

- *Bartlett's Test of Sphericity:*

This test determines whether or not the data is suitable for factor analysis. The test examines the extent to which the correlation matrix departs from orthogonality. If the correlation matrix has unit determinants, it indicates that the variables are not correlated. If the determinant is close to zero then there is a high correlation between the set of variables. If the hypothesis that the correlation matrix is an identity matrix cannot be rejected because the observed significance level is large, then the use of factor analysis should be questioned.

- *Kaiser-Meyer-Olkin (KMO) Test:*

This test statistic is used to test for sampling adequacy. Is a measure of how suited your data is for Factor Analysis. KMO measure indicates whether or not the variables can be grouped into a smaller set of underlying factors. Higher values close to one generally indicate the factor analysis may be useful with the data set.

According to Kaiser and Rice (1974) the following are the guidelines that are put on the values of a KMO results:

Table 1 KMO Guidelines

KMO Measure	**Recommendation**
0.00 - 0.49	Unacceptable.
0.50 - 0.59	Miserable
0.60-0.69	Mediocre
0.70-0.79	Middling
0.80-0.89	Meritorious
0.90-1.00	Marvelous

Even though Kaiser and Rice (1974) proposed the above guideline, according to Zwick and Velicer (1986), the precision of the recommendation of the KMO test depends on the number of indicators underlying a particular factor.

After the extraction phase, the researcher must decide how many constructs to retain for rotation. Factors are rotated to make them more meaningful and easier to interpret. An important difference between them is that they can create factors that are correlated or uncorrelated with each other. There are two types of factor rotation which are oblique rotations (axes are not maintained at 90 degrees) and orthogonal rotations (axes are maintained at 90 degrees). In oblique rotation, the resulting factors will be correlated whereas in orthogonal rotation the rotated factors will remain uncorrelated. There are different methods of rotation and they are Varimax, Quartimax, Equimax, and Promax. Two orthogonal rotation methods are discussed below;

✓ *Varimax Rotation:*

Varimax is the most popular rotational method. Varimax use orthogonal rotations yielding uncorrelated factors. It attempts to minimize the number of variables that have high loadings on different factors. It projects the variable that loads high on particular principal components to keep the factor loadings to a pattern that is easier for interpretation.

✓ *Quartimax Rotation:*

It is an orthogonal alternative which minimizes the number of factors needed to explain each variable. This type of rotation often generates a general factor on which most variables are loaded to a high or medium degree. Such a factor structure is usually not helpful to a specific research purpose (Gorsuch ,1983). Most researchers do not expect a general factor since it is now being done, most research aims on finding out all the factors.

IV. RESULTS PRESENTATION

Findings are presented according to research questions. In addressing each research question the response were organized as follows: the first section provides the results of the descriptive of the variables used in the study. The second section presents the results of the factor analysis.

➢ *Basic Analysis of the Data:*

- *Age of Respondents:*

The age of respondents was important in the study because the financial performance of insurance companies could not be linked to a particular age group. Different age groups buy insurance policies in Ghana. The study considered the following age categories: Below 18, 18-25, 26-32, 33-40, and above. Table 2 describe the distribution of age of the data set.

Table 2 Age Distribution of Respondents

Age of Respondent	Frequency	Percent
Below 18	41	19.5
26-32	72	34.3
33-40	71	33.8
40	26	12.4
Total	210	100.0

Source: Field Survey

From table 2, It was observed that age group 26-32 participated in the study than any other group making a total of 72 representing 34.3%. It was observed that the age group of 40 and above participated the least in the study making a total of 26 representing 12.4%.

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- *Sex of Respondent:*

Table 3 Distribution of Sex of Respondent

Sex	Frequency	Percent
Female	87	41.4
Male	123	58.6
Total	210	100.0

Source: Field Survey

From Table 3 it was observed that 123 males representing 58.6% purchased most insurance policies whiles 87 females constituting 41.4% buys insurance policies.

- *Educational Background:*

The educational background of the respondents was also taken into consideration during the study; this is because it could be a factor influencing the financial performance of insurance companies in Ghana. Table 4.2 describes the distribution of the educational background of respondents.

Table 4 Distribution of Educational Background of the Respondents

Educational Background	Frequency	Percent
No Formal Education	3	1.4
Basic	6	2.9
Secondary	20	9.5
Tertiary	181	86.2
Total	210	100.0

Source: Field Survey

Table 4 showed that the highest number of respondents that took part in the study had tertiary education with a frequency of 181 representing 86.2%. However, the "no formal education" had the least number of respondents.

- *Most Purchased Insurance Product among Respondents:*

The most purchased insurance product was also taken into consideration to reveal what influenced the policyholders to buy insurance. It was also important in the study to reveal generally which type of insurance policy is the most purchased in Ghana. Table 5 shows the distribution of the frequency and percentages of the most purchased insurance policy among respondents.

Table 5 Distribution of the Most Purchased Type of Insurance Policy among Respondents

Type of Insurance Policy	Frequency	Percent
Life Insurance	65	31.0
General Insurance	145	69.0
Total	210	100.0

Source: Field Survey

From Table 5, it was observed that 145 out of 210 policyholders buy General insurance constituting 69% while Life insurance was 65 representing 31%. This suggested that the policyholders buy General insurance more than Life insurance.

- *Distribution of Status of Respondents:*

Table 6 describes the distribution of the status of the respondents on the financial performance of insurance companies. This data was necessary in revealing the number of policy holders in the status category who buys insurance.

Table 6 Distribution of Status among Respondents

Status of Respondent	Frequency	Percent
Government Worker	42	20.0
Private Worker	115	54.8
Self-Employed	25	11.9
Unemployed	28	13.3
Total	210	100.0

Source: Field Survey

From Table 6, it was observed that majority of the private workers with a frequency of 115 buys most insurance policies representing 64.8%. However, government workers with a frequency of 42 constitute 20% of respondents who buys insurance whiles the least number of respondents who buys insurance are self-employed workers with a frequency of 25 representing 11.9%.

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➢ *Reliability Statistics:*

From Appendix E, the Cronbach's alpha value was 0.939 and this implies that there was a level of consistency and reliability in the dataset.

➢ *Descriptive of Statistic of Indicators:*

From Appendix C, Based on the cut-off grand mean of 3.7, it was observed that there were high mean values for some particular indicator variables which are V1 (I stand to gain rather than loss), V2 (The premiums are consistent), V3 (The insurance company has high equity level), V5 (The workers are receptive and friendly), V8 (The insurance company has a lot of branches), V9 (I can access my claims anywhere I am), V11 (I fully understand the insurance pricing policy), V13 (The company has good cash equivalence), V14 (I insure because the claim payment is made on time), This however suggested that they have been rated high by most of the policyholders and therefore influence the financial performance of insurance companies. In addition, there was a set of variables that were easily seen in the table and were to be deliberated on for critical observation. They produced the lowest means and had almost equal standard deviation values. Some are V6 (I insure because of the owner of the insurance company), V30 (I insure with the insurance company because it is closer to my place of residence), V32 (I buy insurance because of an effective advertisement on TV and radio) and V33 (I insure because most of my family insure there). Besides, what makes them more significant is the fact that their standard deviation was around 1.

➢ *Correlation Matrix:*

Appendix B shows a 36 x 36 correlation matrix of the variables under study. It was observed that some of the correlations were low and some were high. However, for this study, a correlation of 0.5 was used as a cut-off, a correlation greater than 0.5 will be considered highly correlated, and below will be considered weakly correlated. The use of the cut-off value will help us to identify the variables that belong to a group. From Appendix B it was observed that the highest positive correlation of 0.756 was obtained from V20 (I insure because the insurance company has good policies) and V23 (I insure because the insurance company is reliable). Since it is a positive value, it means that all those who considered the variable V20 (I insure because the insurance company has good policies) also considered V23 (I insure because the insurance company is reliable) as equally important and vice versa. This implies that those who purchased insurance policies because the insurance company has good policies also considered it because the insurance company is reliable. The next highest correlation matrix of 0.725 was observed between V20 (I insure because the insurance company has good policies) and V21 (I insure because the company policies are well understood). Since it is also a positive value, it implies that those who buy insurance because the insurance company has good policies also buy insurance because the company policies are well understood. A correlation matrix of 0.719 was also observed between V21 (I insure because the company policies are well understood) and V23 (I insure because the insurance company is reliable). It can also be seen that there exists a high pairwise correlation coefficient among some of the indicator variables. Variables with high correlation coefficients greater or equal to 0.5 belong together and can be put into a group. Based on the above the following groups can be identified (V5, V8, V10), (V6, V32, V33), (V9, V10, V20, V21, V23, V29), (V22, V24, V27), and (V30, V32, V33). That is, the variables in each set suggest a group of 5.

Thirty-six indicator variables were involved in this study. The correlations among the indicator variables revealed that five (5) groups could be formed from the 36 indicator variables. However, some of the variables forming the group were correlating multiple times with each other.

➢ *Factor Analysis:*

Factor Analysis was used in the analysis of the sample data. In our basic analysis, the correlation analysis suggested about 4 factors that are best in describing the factors that influence the financial performance of insurance companies in Ghana among the respondents. Hence factor analysis was performed to confirm or otherwise.

Table 7 KMO and Bartlett's Test

Measure	**Value**
Kaiser-Meyer-Olkin Measure of Sampling Adequacy	0.854
Bartlett's Test critical value	5509.106
Bartlett's Test Degree of freedom	630
Bartlett's Test significant value	0.000

Source: Field Survey

Table 7 showed the KMO value of 0.854 is closer to 1 indicating that the test was very sufficient for factoring and will provide a very good result. Also, the Bartlett's Test of Sphericity is also highly significant with the p-value of 0.00, this means that we have enough correlation for factor analysis. These values suggest that factor analysis is suitable to be used for the data set.

➢ *Total Variance Explained:*

In determining how many components to be extracted, the Kaiser's Criterion was used. The Criterion is based on the value of eigen values. The eigen values greater than 1 is used.

Table 8 Total Variance Explained

	Total	% Of Variance	Cumulative %
1	13.098	36.383	36.383
2	4.025	11.182	47.565
3	2.000	5.555	53.120
4	1.627	4.521	57.641
5	1.406	3.907	61.547
6	1.288	3.578	65.126
7	1.020	2.833	67.958
8	1.006	2.796	70.754

Source: Field Survey

It was observed from Table 8 that, going by the eigen values greater-than-one concept eight eigen values were greater than one. It is however noted that two of the eigen values though greater than one is very close to one which is a suspect. There is therefore the need to use the Scree Plot to determine the cut-off loadings.

- *The Scree Plot:*

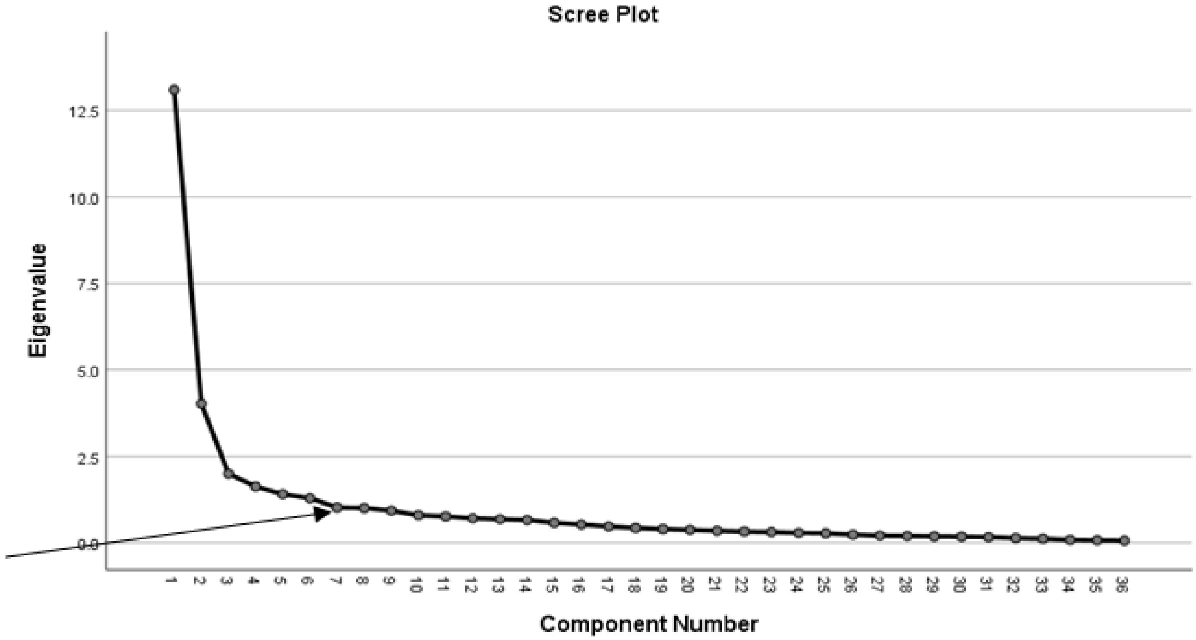

Fig 1 Scree Plot of Eigen Values

The Scree Plot shows that seven main components are very important in explaining the factors associated with the determinant of financial performance of insurance companies in Ghana since it was seen from the diagram that the elbow of the diagram occurs at the seventh component. It further suggests that the number of factors that must be considered should not exceed seven.

- *Factor Rotation:*

After conducting the unrotated factor matrix, some of the loadings were very difficult to interpret hence it was very difficult to assign a label to the factors. Therefore, to obtain a better interpretation of the factors, the components were rotated. To rotate a component for better interpretation, the Varimax rotation was performed and the results are presented in Table 9 Varimax rotation is used to obtain a factor structure in which each variable loads highly on one and only one factor.

Table 10 Varimax Rotated Factor Matrix

	Component						
Variables	1	2	3	4	5	6	7
V1	.333	.179	-.072	.033	.159	-.041	-.134
V2	.073	.152	.208	.104	.018	-.070	.348
V3	-.077	.332	-.100	.091	.219	.219	-.095
V4	.446	.144	-.133	.100	-.051	.409	-.258
V5	**.747**	.007	.007	.253	.065	-.159	.072

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V6	-.045	.042	.707	.039	.371	.171	-.044
V7	.595	.354	.100	.030	.154	.393	.050
V8	**.748**	.083	.020	-.005	.112	.332	.203
V9	.629	.307	-.134	.045	.161	.385	.049
V10	.735	.332	.049	.096	-.011	-.009	.130
V11	.130	**.742**	.099	.237	.083	.080	-.116
V12	.194	.158	.051	.138	.218	.043	.701
V13	.379	**.695**	.082	.090	.172	-.058	.042
V14	.448	.247	.138	.152	.487	-.247	.106
V15	.282	.350	.063	.158	.373	.290	.308
V16	.104	.212	.057	.084	.799	.104	.196
V17	.025	**.769**	.088	-.031	.138	.111	.312
V18	.284	**.709**	.171	.184	.064	.183	-.038
V19	.263	**.663**	-.012	.261	.103	.090	.343
V20	.502	**.609**	.022	.295	-.018	.198	.065
V21	**.516**	.490	-.058	.396	-.128	.069	.028
V22	.222	.288	.235	.354	.093	.658	.133
V23	.572	**.556**	.045	.346	-.010	.071	.094
V24	.305	.234	.265	.635	-.041	.281	.199
V25	.239	.234	.225	.757	.265	.042	.092
V26	.183	-.055	.581	.362	.241	.273	-.029
V27	.289	.468	.265	.619	.175	.117	-.007
V28	**.541**	.158	-.038	.169	-.248	.361	.425
V29	**.629**	.456	-.002	.277	.115	.002	.049
V30	.019	.042	.807	.091	-.016	.107	.080
V31	.153	.359	.402	.516	-.192	.029	.183
V32	-.088	-.005	.823	.145	.059	-.054	-.096
V33	.026	.051	.877	.109	-.036	-.006	.153
V34	.132	.250	.733	-.025	-.176	-.223	.019
V35	.329	**.545**	.043	.031	.069	-.019	.434
V36	**.634**	.208	.291	.214	.111	.017	.070

Source: Field Survey

In Table 10, a cut-off value of 0.5 is used. Thus, loadings greater than or equal to 0.5 will be taken as high and those less than 0.5 will be considered as low. It was observed that factor one loaded highly on V5 (The workers are receptive and friendly), V7 (The insurance company is effective on online media platforms), V8 (The Insurance has a lot of branches), V9 (I can access my claims anywhere I am), V10 (The insurance company has a good policy awareness creation), V21 (I insure because the insurance company policies are well understood), V23 (I insure because the insurance company is reliable), V28 (The insurance company has been in existence for a long time), V29 (The insurance company does well on the financial market) had similar loadings, V36 (The agents have good communicative skills). Therefore, the first factor can be described as an effective management factor.

The second factor has a high significant loading from V11 (I fully understand the insurance pricing policy), V13 (The company has a good cash equivalence), V17 (I insure because of its reputation), V18 (I insure because of its goodwill), V19 (I insure because it is the best insurance company you can invest), V20 (I insure because the insurance company has good policies), V23 (I insure because the insurance company is reliable). Hence, the second factor can be labelled as a competition factor.

The third factor loaded very highly on V6 (I insure because of the owner of the insurance company), V26 (I insure with the insurance company because renowned personalities insure with the company), V30 (I insure with the insurance company because it is closer to my place of residence), V32 (I buy insurance because of an effective advertisement on TV and radio), V33 (I insure because most of my family insure there), V34 (I insure because my friend introduced me to that insurance company). The third factor is therefore labelled as a personality influence factor.

The fourth factor also loads high on V24 (I insure because the insurance company has a good way of managing its losses), V25 (I insure because the company has a good capacity for pooling losses), V27 (I insure because the returns are high), V31 (I buy insurance due to reasonable premium). The fourth factor is, therefore, referred to as the liquidity factor.

The fifth, sixth, and seventh factors were only one-factor indicator and therefore it was not significant enough to assign a factor label to those loadings. Ideally, four factors are clearly established.

➢ *Final Factor Solution:*

After observing the solutions which were obtained from the rotated factor matrices, it provided the salient factors that best explained the variations in factors that influence the financial performance of insurance companies. It is therefore important to know that the varimax transformation is crucial in determining the order of importance of the factors. It was observed from the analysis that a four-factor solution is appropriate and adequate in describing why there are differences in the financial performance of insurance companies in Ghana. The first factor was effective management; the next was competition, followed by personality influence, and finally liquidity.

V. DISCUSSION AND FINDINGS

The research was based on the factors that influence the financial performance of insurance companies in Ghana. The research was conducted to identify the most influential factor that influences financial performance in Ghana and to find out the most purchased insurance policy among policyholders in Ghana. With regards to the most purchased insurance policy, the life insurance policy happens to be the most purchased insurance policy among all respondents having a percentage of 69.0 followed by general insurance having a percentage of 31.

Moreover, the study examined the correlation coefficients between pairs of variables. The highest correlation coefficient 0.756 was observed between the pairs of variables V20 (I insure because the insurance company has good policies) and V23 (I insure because the insurance company is reliable).

The KMO value and the p-value of Bartlett's Test of Sphericity obtained suggested that Factor Analysis Technique was suitable to analyze the data.

After however, conducting the Factor Analysis of the data, the revelations were that the factors that accounted for the financial performance of insurance companies are decreasing in order of importance: effective management, competition, personality influence, and liquidity.

Much research has been conducted on the factors that influence the financial performance of insurance companies, notable among them is the research conducted by Wanjugu (2014), according to her finding profitability influences the financial performance of insurance companies. This was not in line with our research.

The study conducted by M. Al-Dwiry& T. Al Shaher (2020) revealed that Inter develop an insurance culture is the most influential factor which influences the financial performance of insurance companies. This was in line with our research.

Moreover, our study also revealed that the effective management factor was the most influential and it also revealed that profitability was the last factor which suggested it hardly influenced the financial performance of insurance companies.

In addition, the findings of our research revealed that the majority of the policyholders buy General insurance policies. The study revealed that people below the age of 18 purchased few insurance policies and people between the ages of 26 to 32 purchased the most insurance policies.

VI. CONCLUSION AND RECOMMENDATION

- *In this Study, the following Conclusions were Drawn:*
- General Insurance is the most purchased policy among respondents in Ghana
- Four factors were considered important when it comes to the financial performance of insurance companies. The most influential factor is the effective management factor.
- The other factors were competition, personality influence, liquidity, and profitability.

- *The following Recommendations were However made from the Study:*
- There is the need for insurance companies to establish more branches to place of residence of policyholders, create policy awareness, and provide reliable insurance services since effective management factor came up as the most influential factor and it also revealed that most policyholders and customers purchase insurance products when the insurance companies are close to them and does well on the financial market.
- It also suggested that insurance companies should adopt an effective management strategy as a means of ensuring high insurance patronage and penetration in the insurance industry and increasing the capital in companies and insurance premiums. Therefore, insurance companies must take into account the shortage in human resources and try to increase and qualify them and organize them to supervise the insurance process.
- Finally, it is also recommended that policyholders and insurance companies should be sensitized more on the awareness creation of insurance policies since profitability is not the most influential factor.

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APPENDIX - A

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

Table 11 Frequency Distribution of Various Indicators

	Frequency				
Variables	**1**	**2**	**3**	**4**	**5**
V1	10	5	41	70	34
V2	6	5	43	74	82
V3	6	15	29	86	74
V4	16	20	50	124	0
V5	0	13	40	84	73
V6	98	41	21	27	23
V7	17	20	43	65	65
V8	6	16	30	66	92
V9	4	17	26	67	96
V10	1	19	49	71	70
V11	6	13	36	83	72
V12	22	46	33	43	66
V13	0	15	32	80	83
V14	5	18	56	79	52
V15	11	23	55	54	67
V16	44	26	46	42	52
V17	5	18	48	64	75
V18	0	6	61	78	65
V19	3	19	47	55	86
V20	0	18	28	84	80
V21	4	16	36	82	72
V22	10	24	67	60	49
V23	7	18	39	67	79
V24	12	18	59	74	47
V25	9	25	74	35	47
V26	25	46	46	54	39
V27	9	40	57	55	49
V28	4	18	26	64	98
V29	2	7	28	81	91
V30	41	55	37	43	33
V31	6	21	59	70	53
V32	39	46	59	33	32
V33	46	62	59	18	24
V34	31	30	55	44	49
V35	4	10	38	75	82
V36	2	11	47	91	58

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APPENDIX – B

➢ *Correlation Matrix:*

	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V16	V17	V18
000																	
268	1.000																
412	0.357	1.000															
429	0.153	0.366	1.000														
360	0.225	0.075	0.343	1.000													
141	0.157	0.009	-0.024	0.021	1.000												
340	0.209	0.232	0.437	0.482	0.183	1.000											
290	0.227	0.163	0.367	0.532	0.036	0.634	1.000										
287	0.151	0.211	0.449	0.382	-0.024	0.611	0.634	1.000									
263	0.245	0.199	0.410	0.563	0.005	0.526	0.601	0.571	1.000								
322	0.225	0.343	0.352	0.279	0.155	0.442	0.214	0.338	0.381	1.000							
100	0.174	0.011	0.058	0.186	0.095	0.194	0.324	0.253	0.279	0.211	1.000						
395	0.302	0.378	0.257	0.393	0.057	0.477	0.432	0.461	0.540	0.696	0.334	1.000					
291	0.376	0.258	0.269	0.396	0.161	0.347	0.361	0.366	0.522	0.339	0.312	0.474	1.000				
390	0.285	0.302	0.277	0.325	0.096	0.558	0.367	0.457	0.406	0.370	0.308	0.466	0.478	1.000			
369	0.226	0.351	0.150	0.213	0.305	0.338	0.233	0.280	0.203	0.272	0.310	0.366	0.444	0.579	1.000		
184	0.292	0.285	0.126	0.141	0.149	0.408	0.232	0.355	0.318	0.604	0.337	0.550	0.330	0.450	0.367	1.000	
321	0.298	0.360	0.391	0.256	0.126	0.451	0.346	0.432	0.548	0.630	0.232	0.645	0.450	0.519	0.283	0.563	1.000
217	0.344	0.340	0.201	0.311	0.078	0.487	0.423	0.470	0.491	0.498	0.356	0.569	0.381	0.551	0.336	0.601	0.536
379	0.293	0.286	0.401	0.434	0.020	0.584	0.547	0.566	0.545	0.566	0.291	0.708	0.377	0.491	0.296	0.519	0.676
304	0.199	0.193	0.398	0.510	-0.028	0.503	0.392	0.538	0.567	0.511	0.214	0.506	0.358	0.404	0.175	0.392	0.520
107	0.221	0.222	0.331	0.180	0.287	0.528	0.461	0.433	0.315	0.368	0.311	0.320	0.205	0.497	0.261	0.311	0.526
359	0.344	0.247	0.395	0.494	0.054	0.577	0.497	0.603	0.667	0.545	0.231	0.655	0.502	0.526	0.237	0.463	0.636
182	0.283	0.230	0.376	0.396	0.237	0.446	0.414	0.356	0.434	0.451	0.350	0.418	0.330	0.443	0.220	0.341	0.488
216	0.294	0.255	0.180	0.450	0.300	0.424	0.298	0.329	0.343	0.463	0.257	0.415	0.430	0.443	0.406	0.304	0.434
110	0.205	0.122	0.157	0.199	0.478	0.170	0.266	0.175	0.201	0.239	0.189	0.237	0.271	0.283	0.272	0.049	0.264
212	0.232	0.219	0.247	0.304	0.295	0.463	0.317	0.366	0.468	0.481	0.269	0.463	0.404	0.404	0.333	0.346	0.648
171	0.186	0.091	0.370	0.423	-0.068	0.477	0.502	0.479	0.540	0.230	0.335	0.314	0.192	0.447	0.117	0.308	0.377
232	0.208	0.127	0.368	0.465	0.073	0.490	0.531	0.572	0.595	0.414	0.292	0.595	0.494	0.415	0.244	0.385	0.547
134	0.128	-0.007	-0.102	0.075	0.490	0.091	0.071	-0.033	0.063	0.143	0.097	0.135	0.137	0.119	0.075	0.123	0.216
207	0.244	0.116	0.174	0.253	0.200	0.345	0.214	0.206	0.344	0.447	0.263	0.368	0.208	0.345	0.125	0.363	0.400
009	0.182	-0.040	-0.127	-0.013	0.517	0.043	-0.046	-0.113	0.006	0.123	-0.055	0.048	0.173	0.066	0.075	0.107	0.156
045	0.230	-0.093	-0.093	0.047	0.573	0.147	0.059	-0.042	0.116	0.122	0.172	0.141	0.146	0.196	0.117	0.163	0.192
001	0.135	-0.090	-0.088	0.073	0.443	0.127	-0.010	-0.080	0.165	0.150	0.126	0.218	0.152	0.068	-0.026	0.131	0.250
195	0.272	0.245	0.200	0.338	0.050	0.451	0.354	0.351	0.503	0.429	0.395	0.474	0.358	0.410	0.349	0.493	0.478
229	0.083	0.021	0.200	0.464	0.199	0.510	0.499	0.400	0.477	0.274	0.317	0.372	0.360	0.290	0.237	0.215	0.398

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V19	V20	V21	V22	V23	V24	V25	V26	V27	V28	V29	V30	V31	V32	V33	V34	V35	V
1.000																	
0.656	1.000																
0.564	0.725	1.000															
0.439	0.541	0.407	1.000														
0.687	0.756	0.719	0.472	1.000													
0.458	0.595	0.480	0.559	0.564	1.000												
0.473	0.477	0.465	0.474	0.572	0.709	1.000											
0.158	0.269	0.187	0.444	0.293	0.429	0.418	1.000										
0.594	0.602	0.575	0.600	0.653	0.616	0.698	0.469	1.000									
0.391	0.515	0.536	0.411	0.554	0.493	0.321	0.201	0.283	1.000								
0.587	0.646	0.618	0.381	0.714	0.440	0.462	0.269	0.582	0.487	1.000							
0.072	0.088	0.075	0.272	0.155	0.308	0.266	0.565	0.306	0.112	0.027	1.000						
0.454	0.448	0.408	0.432	0.452	0.597	0.560	0.354	0.569	0.293	0.357	0.325	1.000					
-0.004	-0.011	-0.040	0.157	0.019	0.283	0.309	0.510	0.233	-0.111	0.036	0.642	0.367	1.000				
0.118	0.118	0.080	0.250	0.164	0.370	0.324	0.500	0.337	0.065	0.049	0.668	0.475	0.689	1.000			
0.163	0.203	0.138	0.178	0.165	0.203	0.187	0.243	0.298	0.047	0.171	0.473	0.432	0.460	0.649	1.000		
0.599	0.467	0.473	0.283	0.568	0.328	0.294	0.100	0.453	0.429	0.455	0.158	0.358	0.002	0.116	0.136	1.000	
0.351	0.550	0.367	0.413	0.493	0.466	0.425	0.299	0.534	0.355	0.507	0.274	0.440	0.134	0.259	0.390	0.389	

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APPENDIX – C

Table 12 Descriptive Statistics of Indicators

Descriptive Statistics	N	Mean	Std. Deviation
V1	210	**4.01**	1.060
V2	210	**4.05**	.974
V3	210	**3.99**	1.019
V4	210	3.34	.937
V5	210	**4.03**	.888
V6	210	2.22	1.424
V7	210	3.67	1.234
V8	210	**4.06**	1.070
V9	210	**4.11**	1.034
V10	210	**3.90**	.983
V11	210	**3.96**	1.011
V12	210	3.40	1.395
V13	210	**4.10**	.910
V14	210	**3.74**	1.004
V15	210	3.68	1.181
V16	210	3.15	1.463
V17	210	**3.89**	1.065
V18	210	**3.96**	.846
V19	210	**3.96**	1.062
V20	210	**4.08**	.925
V21	210	**3.96**	.997
V22	210	3.54	1.111
V23	210	**3.92**	1.097
V24	210	3.60	1.099
V25	210	3.50	1.095
V26	210	3.17	1.294
V27	210	3.45	1.166
V28	210	**4.11**	1.047
V29	209	**4.21**	.866
V30	209	2.87	1.370
V31	209	3.68	1.050
V32	209	2.87	1.315
V33	209	2.58	1.246
V34	209	3.24	1.355
V35	209	**4.06**	.969
V36	209	**3.92**	.892

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APPENDIX - D

- *Summary of Original Indicators:*
- V1- I stand to gain rather than loss
- V2- The premiums are consistent
- V3- The insurance company has high equity level
- V5- The insurance company is creative and innovative
- V5- The workers are receptive and friendly
- V6- I insure because of the owner of the insurance company
- V7- The insurance company is effective on online media platforms
- V8- The insurance company has a lot of branches
- V9- I can access my claims anywhere I am
- V10- The insurance company has a good policy awareness creation
- V11- I fully understand the insurance pricing policy
- V12- The insurance company charges different premiums for different people
- V13- The company has a good cash equivalence
- V14- I insure because the claim payment is made on time
- V15- I insure because a lot of people have received their claims
- V16- The insurance company rewards me when I don't make a claim
- V17- I insure because of its reputation
- V18- I insure because of its goodwill
- V19- I insure because it is the best insurance company you can invest
- V20- I insure because the insurance company has good policies
- V21- I insure because the company policies are well understood
- V22- I insure with the insurance company because it has a lot of customers
- V23- I insure because the insurance company is reliable
- V24- I insure because the insurance company has a good way of managing its losses
- V25- I insure because the insurance company has a good capacity of pooling losses
- V26- I insure with the insurance company because renowned personalities insure with the Company
- V27- I insure because the returns are high
- V28- The insurance company has been in existence for a long time
- V29- The insurance company does well on the financial market
- V30- I insure with the insurance company because it is closer to my place of residence
- V31- I buy insurance due to reasonable premium
- V32- I buy insurance because of effective advertisement on TV and Radio
- V33- I insure because most of my family insure there
- V34- I insure because my friend introduced me to that insurance company
- V35- The insurance company has a good brand image
- V36- The agent has a good communicative skill

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APPENDIX – E

Table 13 Summary Item Statistics

	Mean	Minimum	Maximum	Range	Maximum / Minimum	Variance	N of Items
Item Means	3.670	2.215	4.206	1.990	1.898	.229	36
Inter-Item Covariances	.367	-.215	1.156	1.370	-5.384	.046	36

Source: Field Survey

Table 14 Reliability Statistics

Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of Items
.939	.944	36

Source: Field Survey